FIDE: Fiber-Informed Differentiable Experimental Design for Transportable Law Reconstruction

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Abstract

Aggregate measurements typically constrain only a small number of expectations of an evolving population and therefore identify a family of compatible probability laws rather than a unique microscopic state. We introduce Fiber-Informed Differentiable Experimental Design (FIDE), which selects measurement systems not only for scientific adequacy but also for the dynamical compatibility of the complete law they imply. For each candidate design, FIDE information-projects a frozen endpoint-informed reference law onto the observed moment fiber and measures the minimum kinetic correction required for the resulting law path to satisfy the continuity equation. We characterize this Full action through a density-weighted Poisson problem and show that its correction decomposes orthogonally into a moment-tangent component and a complementary hidden component invisible to the measured moment rates. FIDE then minimizes Full action only among designs satisfying a prescribed scientific-risk tolerance. In an analytic Gaussian-mixture benchmark, the selected design reduces independent-validation Full action by 34.47% while increasing selection risk by only 0.735%. In a retrospective nonlinear double-gyre benchmark with four aggregate sensors and three independently trained frozen references, the confirmed Full-action reduction increases from 8.28% to 15.71% over the evaluated 0.5%–2% risk allowances, with all reference-wise simultaneous 95% lower bounds positive. In a prospective aggregate-only variant, sensor selection is restricted to endpoint and aggregate predictive information; after repairing the Law baseline, the 0.5% and 1% solutions coincide with Law, while at 2% the Full geometry reduces action by 16.20% under an independently trained post-freeze validation reference, with a strictly negative paired 95% action-difference interval. These results show that scientifically similar measurement systems can induce substantially different law-level transport costs, and that this distinction can remain relevant even when intermediate microscopic target states are unavailable during measurement design.

1 Introduction

Many scientific systems are more naturally described by a time-indexed probability law than by a single deterministic state. Examples include ensembles of particles, cells, molecules, tracers and uncertain physical fields. Yet experiments often expose only a small number of aggregate observables, such as detector intensities, regional occupancies, low-order moments, projections, or pair statistics. Moreover, a finite collection of such expectations generally does not identify the underlying law; instead, it determines an equivalence class of laws consistent with the measurements.

This ambiguity becomes especially consequential when the target is an entire dynamical law. We assume access to an endpoint-informed continuous-time reference dynamics that is trained once and then frozen. Intermediate aggregate measurements provide additional scientific evidence, but different measurement systems can resolve different directions in probability-law space. Consequently, two designs that are nearly indistinguishable according to a scientific reconstruction criterion can imply very different complete law paths and require very different dynamical corrections to the same reference model. This leads to our central question: *among scientifically adequate measurement systems, which induce law reconstructions that are most compatible with a common reference dynamics?* This question remains relevant even when detailed intermediate simulations are available. In the simulation-assisted regime we have in mind, the endpoint information used to construct the frozen reference is independently trusted or separately validated for that purpose, while intermediate simulation is trusted only for particular measurement responses or scientific quantities. Such quantity-specific trust is well established in computational-model validation: model adequacy is assessed for specified quantities of interest, required accuracy, and a domain of intended use, rather than as a blanket property of the simulator (Council et al., 2012). Consequently, access to detailed simulated states does not by itself justify treating the complete simulated intermediate law as physical ground truth or using it as supervision for a full-law neural surrogate.

We introduce *Fiber-Informed Differentiable Experimental Design* (FIDE): for a fixed measurement design, its inner *Moment-Fiber Stochastic Interpolation* (MFSI) construction first lifts the observed moment trajectory to a canonical law-valued path by information projection: at each time, it selects the distribution in the corresponding moment fiber that is closest in relative entropy to the frozen reference marginal. MFSI then measures the minimum L^2 correction to the frozen reference velocity required for this *entire* projected density path to satisfy the continuity equation. We call the resulting minimum kinetic cost the *full-law action*; it quantifies the residual dynamical effort required to realize the measurement-implied law relative to a reference dynamics that has already been learned.

This law-level requirement is substantially stronger than matching only the time derivatives of the measured moments. We show that the minimum correction needed to realize the complete projected law path is characterized by a density-weighted Poisson equation, with an associated action given by a weighted H^{-1} norm of the reference-relative continuity residual. Matching only the selected moment rates yields a weaker finite-dimensional problem: it captures the component of the required correction that is visible through the measured observables, while potentially leaving a non-negative tangent-invisible component unresolved. We characterize exactly when this relaxation is tight and show that, in general, moment-rate compatibility can provide arbitrarily little information about the full law-level transport burden.

Low action alone, however, is not a meaningful experimental-design objective: an uninformative observable may be inexpensive to reconcile with the reference simply because it reveals little beyond what the reference already predicts. FIDE therefore makes *scientific adequacy* primary and uses full-law transportability only to distinguish designs that are already near-optimal under an externally specified scientific risk. A predeclared tolerance determines the admissible degradation in this primary risk, after which FIDE selects the admissible design with minimum full-law action. This ordering avoids an arbitrary scalarization of scientific informativeness and reference-relative dynamical compatibility.

Across two balanced-law benchmarks, FIDE identifies designs with substantially lower held-out full-law realization cost while remaining close to the scientific-risk baseline: the Full design reduces held-out Full action by 34.47% in the analytic Gaussian-mixture benchmark and by 15.71% at the largest evaluated allowance in the nonlinear double-gyre benchmark. In the double gyre, the reduction increases from 8.28% to 15.71% across the evaluated 0.5%–2% risk allowances, and all nine reference-by-allowance simultaneous 95% lower bounds are positive. In the analytic benchmark, a common-discretization audit further shows that approximately 95.5%–97.4% of the Full correction energy of the selected designs lies in the tangent-invisible component. Together, these results show that scientific-risk proximity and moment-rate compatibility do not determine the transport burden of the complete measurement-implied law.

Our main contributions are:

- We formulate aggregate experimental design as a law-identification problem in which measurements define a moment fiber and a frozen learned reference supplies the geometry for a canonical information-projected reconstruction.
- We define full-law transportability through the minimum kinetic correction required to realize the complete projected law path, characterize it by a weighted Poisson equation and weighted H^{-1} norm, and establish an exact Full–Tangent–Hidden decomposition together with sharp conditions separating full-law and moment-tangent transportability.

- We combine this law-level construction with an information-first experimental-design rule and a differentiable finite-sample realization based on empirical information projection, covariance solves, and weighted Poisson, and validate the resulting designs on analytic Gaussian-mixture transport and nonlinear double-gyre advection.

2 Related Work

Experimental design and sensor placement. FIDE builds on optimal experimental design (OED), where measurements are chosen for their value to a downstream task. Bayesian OED casts this choice in decision-theoretic terms (Chaloner & Verdinelli, 1995), with modern methods using variational estimators of expected information gain (Foster et al., 2019) or amortized policies for sequential design (Foster et al., 2021). Goal-oriented OED instead targets uncertainty in a downstream quantity of interest (Attia et al., 2018), while recent control-oriented formulations tailor sensor placement to uncertainty relevant to a subsequent control problem (Madhavan et al., 2026). Sensor-placement methods instantiate these ideas through mutual information (Krause et al., 2008), posterior covariance (Alexanderian et al., 2014), or reconstruction in data-driven low-dimensional representations (Manohar et al., 2018); learned approaches such as PhySense additionally optimize continuous sensor locations using reconstruction feedback (Ma et al., 2025). Robust OED accounts for misspecification of ingredients such as priors and measurement uncertainties (Attia et al., 2025). FIDE shares this task-oriented view but separates scientific adequacy from a secondary notion of dynamical compatibility. Its outer rule has the structure of an ϵ-constraint method (Haimes, 1971): scientific risk defines a near-optimal feasible set, within which FIDE minimizes the reference-relative realization cost of the complete law implied by a design. Thus the contribution is not the generic multiobjective ordering itself, but the law-level secondary criterion to which it is applied.

Aggregate information and entropy-based completion. FIDE operates in a regime where experiments reveal selected aggregate expectations rather than the full microscopic law. Related work studies learning from supervision attached to sets of instances rather than individuals (Zhang et al., 2020), while moment-closure methods construct reduced kinetic descriptions from finitely many moments (Levermore, 1996). In FIDE, finitely many expectations instead define a *moment fiber*: a family of laws consistent with the measurements. Classical maximum-entropy and information-projection principles provide canonical ways to complete partial statistical information (Kullback & Leibler, 1951; Jaynes, 1957; Csiszár, 1975), and recent generative methods can sample maximum-entropy distributions subject to prescribed moments (Lempereur et al., 2026). At the path level, Maximum Caliber extends maximum-entropy inference to distributions over trajectories under dynamical constraints (Dixit et al., 2018), while recent Schrödinger-bridge formulations incorporate ensemble-path information including distribution moments and average currents (Movilla Miangolarra et al., 2024). FIDE differs in that it performs a marginal information projection at each physical time relative to a common frozen reference rather than solving a single path-space entropy problem, and because the constrained observables are themselves the experimental design variables.

Probability dynamics and reference-relative transport. Flow matching and stochastic-interpolant methods learn continuous probability paths and associated velocity fields (Lipman et al., 2023; Albergo & Vanden-Eijnden, 2023; Albergo et al., 2025), with choices such as optimal-transport couplings yielding different and often simpler flows (Tong et al., 2024). Schrödinger bridges instead construct path laws close to a reference process under marginal constraints (De Bortoli et al., 2021), while dynamic-transport methods such as TrajectoryNet infer continuous dynamics from cross-sectional population distributions observed at multiple times (Tong et al., 2020). Other generative approaches impose distributional requirements (Khalafi et al., 2024), average constraints (Hadou et al., 2026), or exact structural constraints through constrained flow geometry (Cheng et al., 2025). FIDE uses these ideas differently: the learned reference dynamics are frozen, and the question is which aggregate constraints should be acquired rather than how to generate samples satisfying a fixed constraint set. Its full-law action inherits the kinetic-energy and continuity-equation viewpoint of dynamic optimal transport (Benamou & Brenier, 2000) and the minimum-norm velocity geometry of Wasserstein tangent spaces (Ambrosio & Serfaty, 2008). Particularly close is optimal transport with prior dynamics, which seeks minimum-energy control steering probability densities through prescribed dynamics (Chen et al., 2016). In FIDE, however, the transport stage does not choose the law path between endpoint marginals: the entire intermediate path has already been fixed by moment-fiber information projection, and the action measures the least correction required to realize that prescribed path relative to a frozen, potentially learned nonlinear reference. This makes full-law transportability a criterion for comparing measurement designs rather than a generative objective in its own right.

3 PROBLEM FORMULATION

We consider a dynamical scientific system whose microscopic state at time $t \in [0, 1]$ is a random variable $X_t \in \mathcal{X} \subseteq \mathbb{R}^d$ with unknown population law P_t. Rather than observing P_t directly, an experiment acquires a finite collection of aggregate measurements intended to retain the information needed for a downstream scientific task.

A measurement design $\boldsymbol{\eta} \in \mathcal{D}$ determines a time-independent observable map and its population measurement trajectory,

$$\boldsymbol{\Phi}_{\boldsymbol{\eta}} : \mathcal{X} \to \mathbb{R}^R, \qquad \boldsymbol{c}_{\boldsymbol{\eta}}(t) := \mathbb{E}_{P_t}[\boldsymbol{\Phi}_{\boldsymbol{\eta}}(X)] \,. \tag{1}$$

The coordinates of $\boldsymbol{\Phi}_{\boldsymbol{\eta}}$ may represent spatial sensors, detector responses, projection angles, localized windows, or coefficients selecting an observable subspace.

Because finitely many expectations generally do not identify P_t, a measurement vector $\boldsymbol{c}$ determines the *moment fiber*

$$\mathcal{F}_{\boldsymbol{\eta}}(\boldsymbol{c}) := \{Q : \mathbb{E}_Q[\boldsymbol{\Phi}_{\boldsymbol{\eta}}(X)] = \boldsymbol{c}\} \,. \tag{2}$$

Thus changing $\boldsymbol{\eta}$ changes not only the measured values but also which directions in probability-law space are constrained and which remain unresolved.

We additionally assume access to a continuous-time reference model trained from endpoint information and frozen before measurement-design optimization. Let $\widetilde{Q}_t$ denote its marginal law, with density $\widetilde{q}_t$, and let $u_t : \mathcal{X} \to \mathbb{R}^d$ satisfy

$$\partial_t \widetilde{q}_t + \nabla \cdot (\widetilde{q}_t u_t) = 0 \tag{3}$$

in the weak sense under suitable boundary or decay conditions. The reference is not assumed to reproduce the unknown intermediate law P_t; it supplies a fixed dynamical geometry against which candidate designs can be compared and is never retrained as $\boldsymbol{\eta}$ changes. In simulation-assisted settings, this distinction permits endpoint information and selected aggregate simulator outputs to be trusted for their stated purpose without treating the complete intermediate simulated law as physical ground truth (Council et al., 2012).

Scientific adequacy is specified independently by an external risk $\mathcal{R}(\boldsymbol{\eta})$. Depending on the setting, this risk may use hidden truth in a controlled benchmark, held-out laws or observables in retrospective data, or predictive information available before acquisition. The design problem therefore provides, for each $\boldsymbol{\eta}$, three ingredients: the aggregate trajectory $\boldsymbol{c}_{\boldsymbol{\eta}}$, the scientific risk $\mathcal{R}(\boldsymbol{\eta})$, and the frozen reference dynamics $(\widetilde{Q}_t, u_t)$.

The central design question is therefore: *among measurement systems that are already scientifically adequate, which imply a complete law evolution that is most dynamically compatible with the fixed reference?* The next section defines the canonical law completion and reference-relative transport cost used by FIDE to answer this question.

4 METHODOLOGY

Given a candidate design $\boldsymbol{\eta}$, FIDE constructs a canonical law path consistent with its aggregate measurements and evaluates the minimum reference-relative correction required to realize that path. We call this inner construction Moment Fiber Stochastic Interpolation (MFSI). Figure 1 summarizes the complete design loop: aggregate measurements define a moment fiber, information projection completes the corresponding law relative to the frozen reference, scientific risk determines admissibility, and Full action measures law-level dynamical compatibility. When the design is continuous, derivatives are propagated through this construction to update the measurement geometry.

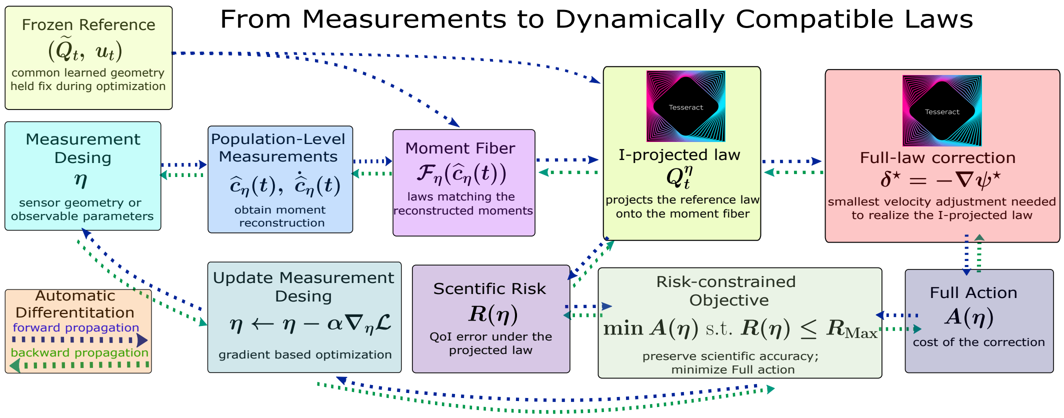

Figure 1: FIDE workflow. A measurement design produces aggregate observations whose reconstructed trajectory defines a moment fiber. The frozen reference is information-projected onto this fiber to obtain the measurement-implied law. Scientific risk evaluates task adequacy, while the Full correction and its action quantify the minimum reference-relative dynamical adjustment required to realize the complete projected law. The risk-constrained objective preserves scientific adequacy while minimizing Full action; for continuous designs, gradients are propagated backward through the pipeline to update the measurement geometry.

Canonical law reconstruction For fixed $(t, \boldsymbol{\eta})$, MFSI selects from the fiber $\mathcal{F}_{\boldsymbol{\eta}}(\boldsymbol{c}_{\boldsymbol{\eta}}(t))$ the law closest in relative entropy to the frozen reference marginal:

$$Q_t^{\boldsymbol{\eta}} \in \arg \min_{Q \ll \widetilde{Q}_t} \left\{ D_{\mathrm{KL}}(Q \| \widetilde{Q}_t) : \mathbb{E}_Q[\boldsymbol{\Phi}_{\boldsymbol{\eta}}] = \boldsymbol{c}_{\boldsymbol{\eta}}(t) \right\}. \tag{4}$$

The resulting path $t \mapsto Q_t^{\boldsymbol{\eta}}$ is the *information-projected law path*. The measurements fix the constrained expectations, while the reference completes the unresolved directions. The reference law and dynamics remain fixed as $\boldsymbol{\eta}$ varies. Section 5 establishes the exponential-tilt representation of $Q_t^{\boldsymbol{\eta}}$ and conditions for uniqueness and differentiability.

Full-law transportability Let $q_t^{\boldsymbol{\eta}}$ denote the density of the projected law. A realizing velocity is written as $u_t + \delta_t$, where δ_t corrects the frozen reference flow. We define the instantaneous Full action by

$$a_{\mathrm{full}}(t, \boldsymbol{\eta}) := \inf_{\delta \in L^2(Q_t^{\boldsymbol{\eta}}; \mathbb{R}^d)} \left\{ \mathbb{E}_{Q_t^{\boldsymbol{\eta}}} \left[\|\delta(X)\|^2 \right] : \partial_t q_t^{\boldsymbol{\eta}} + \nabla \cdot (q_t^{\boldsymbol{\eta}}(u_t + \delta)) = 0 \right\}, \tag{5}$$

and the design-level action by

$$\mathcal{A}(\boldsymbol{\eta}) := \int_0^1 a_{\mathrm{full}}(t, \boldsymbol{\eta})\, \rho(dt), \tag{6}$$

for a fixed finite nonnegative measure ρ on $[0, 1]$. Thus $\mathcal{A}(\boldsymbol{\eta})$ is the least additional kinetic energy required to realize the complete measurement-implied law path relative to the frozen reference.

Let $h_{t,\boldsymbol{\eta}}$ denote the normalized reference-relative continuity residual of the projected path. Under the conditions of Section 5, the unique minimum-energy correction is

$$\delta_t^\star = -\nabla \psi_{t,\boldsymbol{\eta}}^\star, \tag{7}$$

where the mean-zero potential solves

$$\nabla \cdot \left(q_t^{\boldsymbol{\eta}} \nabla \psi_{t,\boldsymbol{\eta}}^\star \right) = q_t^{\boldsymbol{\eta}} h_{t,\boldsymbol{\eta}}, \qquad \mathbb{E}_{Q_t^{\eta}}[\psi_{t,\boldsymbol{\eta}}^\star] = 0. \tag{8}$$

Consequently,

$$a_{\mathrm{full}}(t, \boldsymbol{\eta}) = \mathbb{E}_{Q_t^{\eta}} \left[\left\| \nabla \psi_{t,\boldsymbol{\eta}}^\star \right\|^2 \right]. \tag{9}$$

The numerical experiments evaluate this weighted-Poisson problem using physical spatial discretizations.

Differentiation with respect to measurement design When $\boldsymbol{\eta}$ varies continuously, dependence on the design enters through $\boldsymbol{\Phi}_{\boldsymbol{\eta}}$, the reconstructed moment trajectory $(\boldsymbol{c}_{\boldsymbol{\eta}}, \dot{\boldsymbol{c}}_{\boldsymbol{\eta}})$, the information projection, and the resulting continuity forcing; the reference dynamics remain fixed. The information-projection multiplier is differentiated implicitly through its calibration condition. In particular, when

$$C_{t,\boldsymbol{\eta}} = \mathrm{Cov}_{Q_t^{\boldsymbol{\eta}}}(\boldsymbol{\Phi}_{\boldsymbol{\eta}}, \boldsymbol{\Phi}_{\boldsymbol{\eta}})$$

is nonsingular,

$$D_{\boldsymbol{\eta}}\lambda_{t,\boldsymbol{\eta}}[\xi] = -C_{t,\boldsymbol{\eta}}^{-1} D_{\boldsymbol{\eta}} F[\xi], \tag{10}$$

where F is the moment-calibration residual. Thus differentiation does not require backpropagating through the iterations of the projection solver. Likewise, the discretized Full-action derivative is obtained by implicit differentiation of the gauge-fixed weighted-Poisson system rather than through the iterations of the linear solver. Remaining smooth dependencies are handled by automatic differentiation; Appendix A.10 gives the corresponding sensitivity statement.

Moment-tangent relaxation For comparison, we define a weaker criterion that requires the corrected velocity to reproduce only the time derivatives of the selected measurements. For time-independent $\boldsymbol{\Phi}_{\boldsymbol{\eta}}$,

$$a_{\tan}(t, \boldsymbol{\eta}) := \inf_{\delta \in L^2(Q_t^{\boldsymbol{\eta}}; \mathbb{R}^d)} \left\{ \mathbb{E}_{Q_t^{\boldsymbol{\eta}}}\left[\|\delta(X)\|^2 \right] : \mathbb{E}_{Q_t^{\boldsymbol{\eta}}}\left[J_{\boldsymbol{\Phi}_{\boldsymbol{\eta}}}(u_t + \delta) \right] = \dot{\boldsymbol{c}}_{\boldsymbol{\eta}}(t) \right\}, \tag{11}$$

with

$$\mathcal{A}_{\tan}(\boldsymbol{\eta}) := \int_0^1 a_{\tan}(t, \boldsymbol{\eta})\, \rho(dt).$$

This constrains only the R measured moment rates and is therefore a relaxation of complete law realization. Section 5 derives its finite-dimensional minimum-norm solution and characterizes the component of the Full correction that it cannot detect.

Information-first experimental design Full-law transportability is used only to distinguish designs that are already scientifically adequate. Define the ideal scientific-risk optimum

$$\mathcal{R}^\star := \inf_{\boldsymbol{\eta} \in \mathcal{D}} \mathcal{R}(\boldsymbol{\eta}),$$

and assume $0 < \mathcal{R}^\star < \infty$. For a predeclared tolerance $\varepsilon \geq 0$, let

$$\mathcal{D}_\varepsilon := \{\boldsymbol{\eta} \in \mathcal{D} : \mathcal{R}(\boldsymbol{\eta}) \leq (1+\varepsilon)\mathcal{R}^\star\}, \qquad \boldsymbol{\eta}_{\mathrm{Full}} \in \arg\min_{\boldsymbol{\eta} \in \mathcal{D}_\varepsilon} \mathcal{A}(\boldsymbol{\eta}). \tag{12}$$

Scientific risk therefore defines the admissible set, and Full action ranks only within that set. The corresponding *Law* design minimizes $\mathcal{R}(\boldsymbol{\eta})$ alone, using an action-independent tie-breaking rule. At the idealized population level, this ordering ensures that Full cannot trade arbitrary losses in scientific adequacy for lower transport cost.

In the finite numerical experiments, $\mathcal{R}^\star$ is not assumed to be known exactly. Instead, a numerically optimized Law geometry is frozen before Full selection, and its audited selection risk R_{Law} serves as the operational anchor for the prescribed percentage-risk band. This finite-sample implementation preserves the same information-first ordering while avoiding a claim of global optimality for the numerical Law solution.

5 Theoretical Analysis

We collect only the results needed to characterize MFSI and distinguish Full from moment-tangent transportability. Proofs, regularity conditions, equality cases, counterexamples, and sensitivity results are deferred to Appendix A.

Canonical lift and reference-relative forcing

Theorem 1 (Canonical information-projected lift). *Under the conditions of Appendix A.1, the information projection in equation 4 is unique at the law level and has density*

$$q_t^{\boldsymbol{\eta}}(x) = \widetilde{q}_t(x) \exp\left(\boldsymbol{\lambda}_{t,\boldsymbol{\eta}}^\top \boldsymbol{\Phi}_{\boldsymbol{\eta}}(x) - A_{t,\boldsymbol{\eta}}(\boldsymbol{\lambda}_{t,\boldsymbol{\eta}}) \right), \tag{13}$$

where $A_{t,\boldsymbol{\eta}}(\lambda) = \log \mathbb{E}_{\widetilde{Q}_t}[\exp(\lambda^\top \boldsymbol{\Phi}_{\boldsymbol{\eta}}(X))]$ *and*

$$\mathbb{E}_{Q_t^{\boldsymbol{\eta}}}[\boldsymbol{\Phi}_{\boldsymbol{\eta}}] = \boldsymbol{c}_{\boldsymbol{\eta}}(t). \tag{14}$$

Let $m_{t,\boldsymbol{\eta}} = J_{\boldsymbol{\Phi}_{\boldsymbol{\eta}}} u_t$ and $C_{t,\boldsymbol{\eta}} = \mathrm{Cov}_{Q_t^{\boldsymbol{\eta}}}(\boldsymbol{\Phi}_{\boldsymbol{\eta}}, \boldsymbol{\Phi}_{\boldsymbol{\eta}})$. In a locally identifiable representation, differentiation of the calibration condition gives

$$C_{t,\boldsymbol{\eta}} \dot{\boldsymbol{\lambda}}_{t,\boldsymbol{\eta}} = \dot{\boldsymbol{c}}_{\boldsymbol{\eta}}(t) - \mathbb{E}_{Q_t^{\boldsymbol{\eta}}}[m_{t,\boldsymbol{\eta}}] - \mathrm{Cov}_{Q_t^{\boldsymbol{\eta}}}\left(\boldsymbol{\Phi}_{\boldsymbol{\eta}}, \boldsymbol{\lambda}_{t,\boldsymbol{\eta}}^\top m_{t,\boldsymbol{\eta}}\right). \tag{15}$$

Thus the multiplier dynamics require only an $R \times R$ covariance solve. On the positive support of $q_t^{\boldsymbol{\eta}}$, define the reference-relative continuity residual

$$h_{t,\boldsymbol{\eta}} := \frac{\partial_t q_t^{\boldsymbol{\eta}} + \nabla \cdot (q_t^{\boldsymbol{\eta}} u_t)}{q_t^{\boldsymbol{\eta}}}, \tag{16}$$

equivalently

$$\partial_t q_t^{\boldsymbol{\eta}} + \nabla \cdot (q_t^{\boldsymbol{\eta}} u_t) = q_t^{\boldsymbol{\eta}} h_{t,\boldsymbol{\eta}}. \tag{17}$$

The forcing can be evaluated without differentiating the reference density:

$$h_{t,\boldsymbol{\eta}}(x) = \dot{\boldsymbol{\lambda}}_{t,\boldsymbol{\eta}}^\top (\boldsymbol{\Phi}_{\boldsymbol{\eta}}(x) - \boldsymbol{c}_{\boldsymbol{\eta}}(t)) + \boldsymbol{\lambda}_{t,\boldsymbol{\eta}}^\top \Big(m_{t,\boldsymbol{\eta}}(x) - \mathbb{E}_{Q_t^{\boldsymbol{\eta}}}[m_{t,\boldsymbol{\eta}}] \Big), \tag{18}$$

with

$$\mathbb{E}_{Q_t^{\boldsymbol{\eta}}}[h_{t,\boldsymbol{\eta}}] = 0. \tag{19}$$

Appendices A.2–A.3 give the derivation.

Minimum-energy full-law realization At fixed $(t, \boldsymbol{\eta})$, writing $q = q_t^{\boldsymbol{\eta}}$ and $h = h_{t,\boldsymbol{\eta}}$, a Full correction must satisfy

$$\nabla \cdot (q\delta) = -qh. \tag{20}$$

Theorem 2 (Minimum-energy full correction)**.** *Under the conditions of Appendix A.4, the unique minimum-energy correction is*

$$\delta_t^* = -\nabla \psi_{t,\boldsymbol{\eta}}, \tag{21}$$

where the mean-zero potential solves

$$\nabla \cdot (q_t^{\boldsymbol{\eta}} \nabla \psi_{t,\boldsymbol{\eta}}) = q_t^{\boldsymbol{\eta}} h_{t,\boldsymbol{\eta}}, \qquad \mathbb{E}_{Q_t^{\boldsymbol{\eta}}}[\psi_{t,\boldsymbol{\eta}}] = 0. \tag{22}$$

Consequently,

$$a_{\mathrm{full}}(t, \boldsymbol{\eta}) = \mathbb{E}_{Q_t^{\boldsymbol{\eta}}}\left[\|\nabla \psi_{t,\boldsymbol{\eta}}\|^2 \right]. \tag{23}$$

Equivalently, Full action is the squared weighted H^{-1} norm of the reference-relative continuity residual; see Appendix A.5. It therefore measures compatibility of the complete projected law evolution, not only of selected observables.

Moment-tangent relaxation and hidden action For a sufficiently regular law path transported by v_t and time-independent observables,

$$\frac{d}{dt} \mathbb{E}_{q_t}[\boldsymbol{\Phi}_{\boldsymbol{\eta}}] = \mathbb{E}_{q_t}[J_{\boldsymbol{\Phi}_{\boldsymbol{\eta}}} v_t]. \tag{24}$$

For the projected path, define

$$r_{t,\boldsymbol{\eta}} := \mathbb{E}_{Q_t^{\boldsymbol{\eta}}}[J_{\boldsymbol{\Phi}_{\boldsymbol{\eta}}} u_t] - \dot{\boldsymbol{c}}_{\boldsymbol{\eta}}(t), \tag{25}$$

and

$$G_{t,\boldsymbol{\eta}} := \mathbb{E}_{Q_t^{\boldsymbol{\eta}}}\left[J_{\boldsymbol{\Phi}_{\boldsymbol{\eta}}} J_{\boldsymbol{\Phi}_{\boldsymbol{\eta}}}^\top \right]. \tag{26}$$

Proposition 1 (Minimum-energy tangent correction)**.** *If $r_{t,\boldsymbol{\eta}} \in \mathrm{Range}(G_{t,\boldsymbol{\eta}})$, exact moment-rate matching is feasible and the minimum-energy correction is*

$$\delta_{t,\boldsymbol{\eta}}^{\tan}(x) = -J_{\boldsymbol{\Phi}_{\boldsymbol{\eta}}}(x)^\top G_{t,\boldsymbol{\eta}}^\dagger r_{t,\boldsymbol{\eta}}, \tag{27}$$

with

$$a_{\tan}(t, \boldsymbol{\eta}) = r_{t,\boldsymbol{\eta}}^\top G_{t,\boldsymbol{\eta}}^\dagger r_{t,\boldsymbol{\eta}}. \tag{28}$$

If $r_{t,\boldsymbol{\eta}} \notin \mathrm{Range}(G_{t,\boldsymbol{\eta}})$, exact moment-rate matching is infeasible.

Proposition 2 (Full–Tangent–Hidden decomposition)**.** *Let $\delta^{\mathrm{hid}}_{t,\boldsymbol{\eta}} = \delta^*_t - \delta^{\tan}_{t,\boldsymbol{\eta}}$. Under the conditions of Theorem 2, the hidden component is invisible to the measured moment rates and orthogonal to the minimum tangent correction. Hence*

$$a_{\mathrm{full}}(t,\boldsymbol{\eta}) = a_{\tan}(t,\boldsymbol{\eta}) + a_{\mathrm{hid}}(t,\boldsymbol{\eta}), \qquad a_{\mathrm{hid}}(t,\boldsymbol{\eta}) := \mathbb{E}_{Q_t^{\boldsymbol{\eta}}}\left[\left\|\delta^{\mathrm{hid}}_{t,\boldsymbol{\eta}}\right\|^2\right]. \tag{29}$$

After integration in time,

$$\mathcal{A}(\boldsymbol{\eta}) = \mathcal{A}_{\tan}(\boldsymbol{\eta}) + \mathcal{A}_{\mathrm{hid}}(\boldsymbol{\eta}), \qquad \mathcal{A}_{\mathrm{hid}}(\boldsymbol{\eta}) := \int_0^1 a_{\mathrm{hid}}(t,\boldsymbol{\eta})\,\rho(dt), \tag{30}$$

and therefore

$$a_{\tan}(t,\boldsymbol{\eta}) \le a_{\mathrm{full}}(t,\boldsymbol{\eta}), \qquad \mathcal{A}_{\tan}(\boldsymbol{\eta}) \le \mathcal{A}(\boldsymbol{\eta}). \tag{31}$$

The inequality can be strict by an arbitrarily large factor. In particular:

Theorem 3 (No universal tangent control of full-law action)**.** *There is no universal finite constant K such that*

$$a_{\mathrm{full}}(t,\boldsymbol{\eta}) \le K\,a_{\tan}(t,\boldsymbol{\eta}) \tag{32}$$

for all regular MFSI instances; the construction in Appendix A.9 gives a smooth family for which

$$\frac{a_{\mathrm{full}}(t,\boldsymbol{\eta})}{a_{\tan}(t,\boldsymbol{\eta})} \to \infty. \tag{33}$$

Thus the Tangent criterion measures only the component of the required correction visible through the selected moment rates. The hidden-action fraction is defined in Appendix A.7, exact Full–Tangent equality conditions are given in Appendix A.8, and design-sensitivity conditions are given in Appendix A.10.

6 Experiments

We evaluate FIDE in three controlled measurement-design protocols on two transport systems, spanning analytic transport, nonlinear advection, and a prospective aggregate-only design setting. The experiments test a common claim: *measurement systems with nearly identical scientific adequacy can nevertheless induce substantially different law-level transport costs relative to a fixed reference geometry*. The first two protocols test this claim retrospectively with hidden benchmark-population information available during selection, while the third asks whether the same design principle can transfer when selection is restricted to endpoint information and aggregate predictive quantities. We retain only the experimental logic and principal results here and defer the physical models, observation procedures, numerical solvers, sensor geometries, information boundaries, and certification details to Appendix B.

Common experimental pipeline. Across all three protocols, endpoint information is used to train reference dynamics that are frozen during measurement-design optimization, and every candidate geometry is compared against the same frozen reference model or reference ensemble within a given design stage. A candidate geometry determines sparse noisy aggregate sensor information and a smooth reconstructed moment trajectory $\widehat{\boldsymbol{c}}_{\boldsymbol{\eta}}(t)$. In the analytic and retrospective double-gyre benchmarks these quantities are generated from hidden benchmark populations; in the prospective double-gyre protocol, selection instead uses only frozen aggregate response statistics, their finite-sampling covariance, and aggregate scientific-QoI predictions. At each time, the corresponding frozen reference marginal is information-projected onto the reconstructed moments to obtain the measurement-implied law $Q_t^{\boldsymbol{\eta}}$. Scientific risk determines whether the design is admissible, while Full action measures the minimum correction required for the *entire* projected law path to evolve consistently relative to the frozen reference. A numerically optimized *Law* geometry provides the frozen finite-data risk anchor R_{Law}, and action-based designs must satisfy

$$R(\boldsymbol{\eta}) \le \left(1 + \frac{p}{100}\right) R_{\mathrm{Law}}. \tag{34}$$

The Full design minimizes full-law action within this admissible set. Where reported, Tangent minimizes the weaker moment-rate action under the same scientific restrictions. The first two protocols use disjoint frozen selection and evaluation banks. In the prospective protocol, the repaired Law and Full frontier is frozen against the D0 design reference before the independent E1 validation reference and hidden validation trials are generated.

Study	Allowance	Sel. risk inc.	Law eval. action	Full eval. action	Reduction
Gaussian-mixture transport	4%	0.735%	29.014	19.013	34.47%
Double-gyre tracers	2%	1.556%	1.5915	1.3414	15.71%
Prospective double gyre	2%	1.524%	1.7030	1.4270	16.20%

Table 1: Representative independently evaluated operating points. “Sel. risk inc.” is the Law-relative increase in finite-data scientific risk on the selection data. Evaluation uses the independent validation bank for the Gaussian-mixture benchmark and the independent holdout for the retrospective double-gyre benchmark. In the prospective benchmark, D0 denotes the frozen endpoint-trained reference used for sensor selection, whereas E1 denotes the independent endpoint-trained reference created only after the complete sensor-design frontier was frozen and used exclusively for held-out validation. The prospective 2% selection-risk increase is therefore computed relative to Law under D0, while its held-out E1 risk increase is 1.289%. Action magnitudes are benchmark- and reference-specific and should be compared only within a protocol.

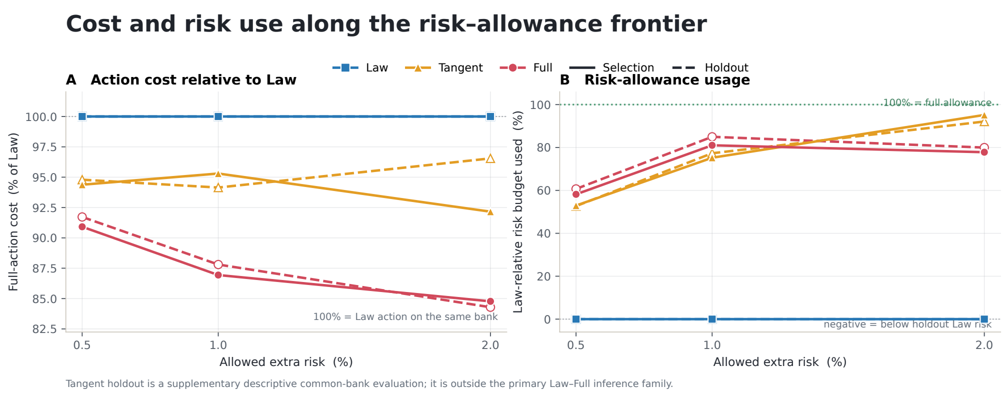

Figure 2: Risk-controlled comparison in the retrospective double-gyre benchmark. Full has lower common Full action than Law and Tangent at every evaluated allowance on both selection and independent holdout data. Solid curves denote selection quantities and dashed curves denote holdout cross-evaluations; only selection risk is constrained by the prescribed allowance.

Analytic Gaussian-mixture transport. The first benchmark isolates the design effect in a setting where the hidden population path is analytically available to the benchmarker. A two-dimensional Gaussian mixture evolves from horizontally separated lobes to vertically separated lobes through an uncertain intermediate orientation, while two localized Gaussian sensors are constrained to a radius-1.5 ring. Each finite trial observes 100 particles at 11 acquisition times with additive detector noise, and the reference flow is trained from the common endpoint distributions only. At the 4% allowance, the Full geometry uses only a 0.735% increase in selection risk and reduces independent-validation Full action from 29.014 to 19.013, a 34.47% reduction. The common-raster Full–Tangent–Hidden audit gives $\Gamma_h \approx 0.955$–0.974 for the Full-selected geometries, so most of the required law-level correction is invisible to the two measured moment-rate directions. Complete results are given in Appendix B.2.

Double-gyre passive tracers. The second benchmark tests FIDE under nonlinear, time-dependent advection on $[0, 2] \times [0, 1]$. Four Gaussian sensors observe $2{,}000$ particles at nine acquisition times, while three independently trained endpoint-only references provide the common reference geometry. Over the confirmed allowances $p \in \{0.5, 1, 2\}$, Full reduces independent-holdout action relative to Law by 8.28%, 12.19%, and 15.71%, respectively. At 2%, the selected design incurs a 1.556% increase in selection risk and lowers holdout Full action from 1.5915 to 1.3414. All nine reference-by-allowance simultaneous 95% lower bounds for the Full-versus-Law reduction are strictly positive. Tangent reduces holdout Full action by 5.21%, 5.85%, and 3.45% at the same operating points. This comparison is descriptive and is not part of the primary simultaneous Law–Full inference family. Complete results are given in Appendix B.3.

Prospective aggregate-only double gyre. The third protocol keeps the same bounded double gyre, four Gaussian sensors, finite observation model, and reflected Full-action construction, but changes the information available before selection. The design stage receives only the endpoint ensembles, aggregate sensor-response fields, the complete finite-sampling covariance of the four observations, and aggregate scientific-QoI predictions; no intermediate target particles or hidden simulator path are available. A single D0 endpoint-trained reference is used for selection. After a stronger D0-only audit revealed that the original Law baseline was underoptimized, the Law anchor was repaired to $R_{\mathrm{Law}}^{\mathrm{D0}} = 1.19657794$, Full was rerun at 0.5% and 1%, and a previously audited feasible finalist was retained at 2%. The two tight allowances select the Law geometry itself and therefore yield no action benefit. At 2%, the selected Full geometry has D0 risk 1.21481697, a 1.524% increase over Law and below the 2% ceiling. Only after the complete three-point frontier was frozen was an independent E1 reference trained and 64 fresh paired hidden-validation trials generated. On E1, Full lowers action from 1.70295 to 1.42700, a 16.20% reduction, while the observed scientific-risk increase is 1.2886%. The paired 95% confidence interval for the trial-wise Full-minus-Law action difference is $[-0.31510, -0.23681]$. Tangent is omitted from the repaired prospective comparison because it was not rerun after the Law repair. This experiment therefore provides a prospective transfer test under an aggregate-only selection boundary, but its uncertainty claim is conditional on one D0 design-reference seed and one independent E1 validation-reference seed rather than a multi-reference robustness statement. Complete details are given in Appendix B.4.

Empirical conclusion. The three protocols support a common but deliberately qualified conclusion. The analytic and retrospective double-gyre benchmarks show that proximity in scientific risk does not determine reference-relative dynamical compatibility: designs within tight Law-relative risk neighborhoods can induce substantially different Full action for the complete law path they imply. The analytic benchmark further shows directly that this discrepancy can be dominated by correction directions invisible to the measured moment rates, while the retrospective double gyre confirms that optimizing the Tangent criterion does not recover the Full-law ranking. The prospective experiment extends the evidence to an aggregate-only pre-selection information boundary: no certified gain appears at the repaired 0.5% and 1% allowances, but a distinct 2% design transfers to a post-freeze reference with a 16.20% held-out action reduction. Thus Full-law transportability can provide design information beyond scientific risk even without access to intermediate microscopic target states during selection, although the magnitude and onset of that benefit depend on the admissible risk set and the reference regime.

7 Conclusion

FIDE treats aggregate experimental design as a law-identification problem. Measurements define a moment fiber, a frozen reference supplies a canonical information-projected completion, and Full action quantifies the minimum dynamical correction required to realize the resulting law path. In the balanced setting, this correction is characterized by a density-weighted Poisson equation and decomposes into tangent-visible and tangent-invisible components, making explicit why matching measured moment rates is generally weaker than ensuring dynamical compatibility of the complete inferred law.

The resulting design principle is information-first: scientific risk defines the admissible measurement systems, and Full action ranks only within that set. In the analytic Gaussian-mixture benchmark, FIDE reduces independent-validation Full action by 34.47%. In the retrospective double-gyre benchmark, it achieves confirmed reductions of 8.28%, 12.19%, and 15.71% across the evaluated 0.5%, 1%, and 2% allowances, with positive simultaneous 95% lower bounds under all three frozen references. The prospective aggregate-only double-gyre protocol gives a more qualified result after repairing the numerical Law anchor: the 0.5% and 1% solutions coincide with Law, while the distinct 2% Full geometry reduces post-freeze E1 action from 1.70295 to 1.42700, or 16.20%, with an E1 risk increase of 1.2886% and a paired Full-minus-Law action-difference 95% confidence interval of $[-0.31510, -0.23681]$. Because this prospective validation uses one independent E1 reference, it establishes transfer across the enforced selection–validation boundary but not multi-reference robustness. Taken together, the experiments show that scientifically similar measurement systems can induce materially different law-level transport costs relative to a fixed reference geometry, and that this distinction can remain useful even when sensor selection is restricted to endpoint and aggregate predictive information rather than intermediate microscopic target states.

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A TECHNICAL STATEMENTS AND PROOFS

This appendix collects the technical statements and proofs supporting Section 5. We record the regularity assumptions, exact equality conditions, counterexample construction, design-sensitivity result, and finite-measure unbalanced analogues of the principal balanced statements. All continuity equations are understood weakly unless the displayed densities and vector fields are regular enough for the corresponding strong form. Boundary terms are assumed to vanish under periodic, decay, or compatible no-flux conditions.

For compactness, when $(t, \boldsymbol{\eta})$ is fixed we write

$$\widetilde{q} = \widetilde{q}_t, \qquad q = q_t^{\boldsymbol{\eta}}, \qquad \phi = \boldsymbol{\Phi}_{\boldsymbol{\eta}}, \qquad c = \boldsymbol{c}_{\boldsymbol{\eta}}(t), \qquad \lambda = \boldsymbol{\lambda}_{t,\boldsymbol{\eta}}, \qquad u = u_t. \tag{35}$$

Expectations without an explicit subscript in a proof are taken under the law indicated by the surrounding expression. We repeatedly use four definitions from the main text. The normalized continuity residual is determined by equation 16; a full-law correction therefore satisfies equation 20. For time-independent observables, equation 24 gives the moment-rate constraint, with tangent residual $r_{t,\boldsymbol{\eta}}$ and Gram matrix $G_{t,\boldsymbol{\eta}}$ defined in equation 25–equation 26. These definitions fix the signs used throughout the balanced proofs.

Finite-measure unbalanced extension used below. The unbalanced corollaries use the finite-measure factorization

$$\mu_t(dx) = M_t q_t(x)\, dx, \qquad M_t > 0, \qquad \int q_t(x)\, dx = 1. \tag{36}$$

The normalized shape q_t is information-projected relative to the frozen normalized reference shape $\widetilde{q}_t$, while the finite mass M_t is treated separately. We assume that the frozen reference shape satisfies

$$\partial_t \widetilde{q}_t + \nabla\cdot (\widetilde{q}_t u_t) = 0 \tag{37}$$

and that the reference finite-mass schedule has a spatially constant logarithmic source rate $g_{\mathrm{ref}}(t)$. For a time-independent observable vector ϕ, write

$$\bar{c}_t := \int \phi\, d\mu_t = M_t \mathbb{E}_{q_t}[\phi], \qquad c_t := \frac{\bar{c}_t}{M_t} = \mathbb{E}_{q_t}[\phi]. \tag{38}$$

A corrected finite-measure dynamics is taken to satisfy

$$\partial_t (M_t q_t) + \nabla\cdot (M_t q_t (u_t + \delta_t)) = M_t q_t \left(g_{\mathrm{ref}} + \alpha_t\right), \tag{39}$$

where δ_t is a transport correction and α_t is a reaction-rate correction. Defining

$$h_{\mathrm{shape}} := \frac{\partial_t q + \nabla\cdot (qu)}{q}, \qquad h_{\mathrm{ub}} := h_{\mathrm{shape}} + \frac{\dot{M}}{M} - g_{\mathrm{ref}}, \tag{40}$$

equation equation 39 is equivalent to

$$-\nabla\cdot (q\delta) + q\alpha = q h_{\mathrm{ub}}. \tag{41}$$

For a fixed reaction penalty $\kappa > 0$, the corresponding instantaneous Full action is

$$a_{\mathrm{full}}^{\mathrm{ub}}(t, \boldsymbol{\eta}) := M_t \inf_{\delta,\alpha} \left\{ \mathbb{E}_{q_t}\left[\|\delta\|^2 + \kappa\alpha^2\right] : -\nabla\cdot (q_t\delta) + q_t\alpha = q_t h_{\mathrm{ub}} \right\}. \tag{42}$$

Remark 1 (Mass is separate from the normalized-shape exponential family)**.** *The exact mass M_t should be treated separately from the normalized-shape information projection. In particular, appending the constant observable* 1 *to the shape observable vector makes the covariance matrix $C_{t,\boldsymbol{\eta}}$ singular and is unnecessary because normalization already fixes $\mathbb{E}_q[1] = 1$. A constant observable may nevertheless be appended to the* unbalanced tangent *observable vector after shape calibration; there it represents the raw finite-measure mass and makes the mass-rate equation explicitly visible to the tangent criterion.*

A.1 CANONICAL INFORMATION PROJECTION

Technical conditions. Fix $(t, \boldsymbol{\eta})$. Assume that $\boldsymbol{c}_{\boldsymbol{\eta}}(t)$ lies in the relative interior of the feasible moment set, that the dual objective $A_{t,\boldsymbol{\eta}}(\lambda)-\lambda^\top \boldsymbol{c}_{\boldsymbol{\eta}}(t)$ attains a minimizer $\boldsymbol{\lambda}_{t,\boldsymbol{\eta}}$, and that $A_{t,\boldsymbol{\eta}}$ is finite on an open neighborhood of that minimizer. If $C_{t,\boldsymbol{\eta}} \succ 0$ in a minimal observable representation, the calibrated multiplier is locally unique; without this rank condition, the projected law remains unique but its coordinates need not be.

Proof of Theorem 1. Let

$$\mathcal{Q}_c := \left\{ Q \ll \widetilde{Q}_t : \mathbb{E}_Q[\phi] = c \right\} \tag{43}$$

with the implicit finiteness conditions from equation 4. For any λ at which $A_{t,\boldsymbol{\eta}}(\lambda) < \infty$, define the exponentially tilted law

$$\frac{dQ_\lambda}{d\widetilde{Q}_t}(x) = \exp\bigl(\lambda^\top \phi(x) - A_{t,\boldsymbol{\eta}}(\lambda)\bigr) . \tag{44}$$

The normalization follows directly from the definition of $A_{t,\boldsymbol{\eta}}$.

By assumption, the dual objective

$$D(\lambda) := A_{t,\boldsymbol{\eta}}(\lambda) - \lambda^\top c \tag{45}$$

attains its minimum at $\lambda = \boldsymbol{\lambda}_{t,\boldsymbol{\eta}}$, and $A_{t,\boldsymbol{\eta}}$ is finite on an open neighborhood of that point. On this neighborhood the standard differentiation formula for a log moment-generating function gives

$$\nabla_\lambda A_{t,\boldsymbol{\eta}}(\lambda) = \mathbb{E}_{Q_\lambda}[\phi], \tag{46}$$

and

$$\nabla^2_\lambda A_{t,\boldsymbol{\eta}}(\lambda) = \mathrm{Cov}_{Q_\lambda}(\phi, \phi). \tag{47}$$

Since D is differentiable at an unconstrained minimizer,

$$0 = \nabla D(\boldsymbol{\lambda}_{t,\boldsymbol{\eta}}) = \mathbb{E}_{Q_{\boldsymbol{\lambda}_{t,\boldsymbol{\eta}}}}[\phi] - c. \tag{48}$$

Hence the tilted law $Q_{\boldsymbol{\lambda}_{t,\boldsymbol{\eta}}}$ belongs to $\mathcal{Q}_c$ and satisfies the calibration condition equation 14.

Set $Q^* = Q_{\boldsymbol{\lambda}_{t,\boldsymbol{\eta}}}$. For any feasible $Q \in \mathcal{Q}_c$ for which the displayed relative entropies are finite, the Radon–Nikodym chain rule and equation 44 give

$$\begin{aligned} D_{\mathrm{KL}}(Q\|\widetilde{Q}_t) &= D_{\mathrm{KL}}(Q\|Q^*) + \mathbb{E}_Q\left[\log \frac{dQ^*}{d\widetilde{Q}_t}\right] \\ &= D_{\mathrm{KL}}(Q\|Q^*) + \boldsymbol{\lambda}_{t,\boldsymbol{\eta}}^\top \mathbb{E}_Q[\phi] - A_{t,\boldsymbol{\eta}}(\boldsymbol{\lambda}_{t,\boldsymbol{\eta}}) \\ &= D_{\mathrm{KL}}(Q\|Q^*) + \boldsymbol{\lambda}_{t,\boldsymbol{\eta}}^\top c - A_{t,\boldsymbol{\eta}}(\boldsymbol{\lambda}_{t,\boldsymbol{\eta}}). \end{aligned} \tag{49}$$

For $Q = Q^*$ the final two terms equal $D_{\mathrm{KL}}(Q^*\|\widetilde{Q}_t)$. Therefore

$$D_{\mathrm{KL}}(Q\|\widetilde{Q}_t) - D_{\mathrm{KL}}(Q^*\|\widetilde{Q}_t) = D_{\mathrm{KL}}(Q\|Q^*) \geq 0. \tag{50}$$

Equality holds only when $Q = Q^*$ almost everywhere. Thus the I-projection is unique at the law level and has the density in equation 13.

Finally, equation 47 evaluated at the calibrated multiplier gives

$$\nabla^2_\lambda A_{t,\boldsymbol{\eta}}(\boldsymbol{\lambda}_{t,\boldsymbol{\eta}}) = C_{t,\boldsymbol{\eta}}. \tag{51}$$

If $C_{t,\boldsymbol{\eta}}$ is positive definite in a minimal observable representation, the dual Hessian is nonsingular at the optimizer. By continuity of the Hessian, the dual objective is strictly convex on a sufficiently small neighborhood, and the multiplier is locally unique. If the observable coordinates are redundant, different multipliers can generate the same log-density up to an additive constant on the support of $\widetilde{Q}_t$, so multiplier non-uniqueness does not contradict law-level uniqueness. □

Corollary 1 (Finite-measure information projection with prescribed mass)**.** *Fix $M > 0$ and a raw finite-measure moment vector $\bar{c}$. Let $c = \bar{c}/M$, and suppose the conditions of Theorem 1 hold for this normalized moment vector. Then the finite measure $\mu^* = MQ^*$, where Q^* is the information projection in Theorem 1, is the unique finite measure of mass M whose normalized shape minimizes $D_{\mathrm{KL}}(Q\|\widetilde{Q}_t)$ subject to $\int \phi\, d(MQ) = \bar{c}$. Equivalently, μ^* uniquely minimizes $MD_{\mathrm{KL}}(Q\|\widetilde{Q}_t)$ over the same fixed-mass fiber, and its normalized density has the same exponential-tilt representation equation 13.*

Proof. Every finite measure μ of mass M can be written uniquely as $\mu = MQ$ with Q a probability law, and the raw constraint $\int \phi \, d\mu = \bar{c}$ is equivalent to $\mathbb{E}_Q[\phi] = \bar{c}/M = c$. Since multiplication of the objective by the positive constant M does not change its minimizer, Theorem 1 applies directly to the normalized shape. $\square$

A.2 Physical-time multiplier dynamics

Proposition 3 (Multiplier dynamics)**.** *Under the regularity assumptions stated below and in a minimal observable representation with $C_{t,\boldsymbol{\eta}} \succ 0$, the calibrated multiplier satisfies equation 15.*

Technical conditions. Fix $\boldsymbol{\eta}$ and a time t. Assume that $\boldsymbol{c}_{\boldsymbol{\eta}}(s)$ is continuously differentiable in s near t, that the reference density and velocity are regular enough for the reference continuity equation and the integrations by parts below, and that the normalized exponential family $q_{\lambda,s}$ defined below is continuously differentiable in (λ, s) near $(\boldsymbol{\lambda}_{t,\boldsymbol{\eta}}, t)$ with differentiation and integration interchangeable. In a minimal observable representation, assume $C_{t,\boldsymbol{\eta}} \succ 0$.

Proof of Proposition 3. Fix $\boldsymbol{\eta}$ and suppress it from the notation. For λ and t in the neighborhood specified above, define

$$q_{\lambda,t}(x) := \widetilde{q}_t(x) \exp\bigl(\lambda^\top \phi(x) - A_t(\lambda)\bigr)\,, \tag{52}$$

and the calibration residual

$$\mathcal{F}(\lambda, t) := \mathbb{E}_{q_{\lambda,t}}[\phi] - c_t. \tag{53}$$

By the stated regularity assumptions, $\mathcal{F}$ is continuously differentiable near (λ_t, t), and

$$\partial_\lambda \mathcal{F}(\lambda_t, t) = \mathrm{Cov}_{Q_t^{\boldsymbol{\eta}}}(\phi, \phi) = C_{t,\boldsymbol{\eta}}. \tag{54}$$

Since $C_{t,\boldsymbol{\eta}} \succ 0$, the implicit-function theorem gives a locally unique continuously differentiable calibrated path $t \mapsto \lambda_t$ satisfying $\mathcal{F}(\lambda_t, t) = 0$. In particular, $\dot{\lambda}_t$ exists, so the differentiations below are justified. Let

$$s_t(x) := \partial_t \log \widetilde{q}_t(x) \tag{55}$$

where the reference density is positive. From the exponential representation,

$$\log q_t(x) = \log \widetilde{q}_t(x) + \lambda_t^\top \phi(x) - A_t(\lambda_t). \tag{56}$$

At fixed λ, differentiation of the log-partition function in physical time gives

$$\partial_t A_t(\lambda) = \mathbb{E}_{q_{\lambda,t}}[s_t]. \tag{57}$$

Along the calibrated path $t \mapsto \lambda_t$,

$$\frac{d}{dt} A_t(\lambda_t) = \mathbb{E}_{Q_t^{\boldsymbol{\eta}}}[s_t] + \dot{\lambda}_t^\top c_t, \tag{58}$$

where $c_t = \mathbb{E}_{Q_t^{\boldsymbol{\eta}}}[\phi]$.

Differentiating the calibration identity $c_t = \mathbb{E}_{Q_t^{\boldsymbol{\eta}}}[\phi]$ and using that ϕ has no explicit physical-time dependence yields

$$\begin{aligned} \dot{c}_t &= \mathbb{E}_{Q_t^{\boldsymbol{\eta}}}[\phi \, \partial_t \log q_t] \\ &= \mathrm{Cov}_{Q_t^{\boldsymbol{\eta}}}(\phi, s_t) + C_{t,\boldsymbol{\eta}} \dot{\lambda}_t. \end{aligned} \tag{59}$$

It remains to express the covariance with s_t without differentiating the reference density explicitly.

Let

$$w_t(x) := \frac{q_t(x)}{\widetilde{q}_t(x)} = \exp\bigl(\lambda_t^\top \phi(x) - A_t(\lambda_t)\bigr)\,. \tag{60}$$

The reference continuity equation gives $\partial_t \widetilde{q}_t = -\nabla \cdot (\widetilde{q}_t u_t)$. Therefore, using the assumed boundary conditions,

$$\begin{aligned} \mathbb{E}_{Q_t^{\boldsymbol{\eta}}}[s_t] &= \int w_t \, \partial_t \widetilde{q}_t \, dx \\ &= -\int w_t \, \nabla \cdot (\widetilde{q}_t u_t) \, dx \\ &= \int \widetilde{q}_t u_t \cdot \nabla w_t \, dx. \end{aligned} \tag{61}$$

Since

$$\nabla w_t = w_t J_\phi^\top \lambda_t, \tag{62}$$

we obtain

$$\mathbb{E}_{Q_t^{\boldsymbol{\eta}}}[s_t] = \mathbb{E}_{Q_t^{\boldsymbol{\eta}}}\left[\lambda_t^\top J_\phi u_t\right] = \mathbb{E}_{Q_t^{\boldsymbol{\eta}}}[\lambda_t^\top m_t]. \tag{63}$$

Similarly, componentwise integration by parts gives

$$\begin{aligned}\mathbb{E}_{Q_t^{\boldsymbol{\eta}}}[\phi s_t] &= \int w_t \phi\, \partial_t \widetilde{q}_t\, dx \\ &= \int \widetilde{q}_t u_t \cdot \nabla(w_t \phi)\, dx \\ &= \mathbb{E}_{Q_t^{\boldsymbol{\eta}}}[m_t] + \mathbb{E}_{Q_t^{\boldsymbol{\eta}}}[\phi\, \lambda_t^\top m_t]. \end{aligned} \tag{64}$$

Subtracting $\mathbb{E}[\phi]\mathbb{E}[s_t] = c_t \mathbb{E}[\lambda_t^\top m_t]$ yields

$$\mathrm{Cov}_{Q_t^{\boldsymbol{\eta}}}(\phi, s_t) = \mathbb{E}_{Q_t^{\boldsymbol{\eta}}}[m_t] + \mathrm{Cov}_{Q_t^{\boldsymbol{\eta}}}(\phi, \lambda_t^\top m_t). \tag{65}$$

Substitution into equation 59 gives

$$C_{t,\boldsymbol{\eta}} \dot{\lambda}_t = \dot{c}_t - \mathbb{E}_{Q_t^{\boldsymbol{\eta}}}[m_t] - \mathrm{Cov}_{Q_t^{\boldsymbol{\eta}}}(\phi, \lambda_t^\top m_t), \tag{66}$$

which is equation 15. Positive definiteness of $C_{t,\boldsymbol{\eta}}$ in the stated minimal representation makes this a uniquely solvable $R \times R$ linear system. ☐

Corollary 2 (Multiplier dynamics for a finite-measure path)**.** *Suppose the assumptions of Proposition 3 hold for the normalized shape q_t, and suppose $M_t > 0$ and the raw moment path $\bar{c}_t$ in equation 38 are continuously differentiable. Then the same covariance equation holds with the normalized target*

$$c_t = \frac{\bar{c}_t}{M_t}, \qquad \dot{c}_t = \frac{\dot{\bar{c}}_t}{M_t} - \frac{\dot{M}_t}{M_t} c_t, \tag{67}$$

namely

$$C_{t,\boldsymbol{\eta}} \dot{\lambda}_{t,\boldsymbol{\eta}} = \frac{\dot{\bar{c}}_t}{M_t} - \frac{\dot{M}_t}{M_t} c_t - \mathbb{E}_{Q_t^{\boldsymbol{\eta}}}[m_{t,\boldsymbol{\eta}}] - \mathrm{Cov}_{Q_t^{\boldsymbol{\eta}}}\left(\boldsymbol{\Phi}_{\boldsymbol{\eta}}, \boldsymbol{\lambda}_{t,\boldsymbol{\eta}}^\top m_{t,\boldsymbol{\eta}}\right). \tag{68}$$

Proof. The information projection acts only on the normalized shape, so Proposition 3 applies with target $c_t = \bar{c}_t / M_t$. Equation equation 67 is the quotient rule, and substitution into equation 15 gives equation 68. ☐

A.3 Density-free continuity forcing

Proposition 4 (Density-free continuity forcing)**.** *Under the assumptions of Proposition 3, the residual defined by equation 16 satisfies equation 18 and equation 19.*

Proof of Proposition 4. Again fix $\boldsymbol{\eta}$ and write $q_t = q_t^{\boldsymbol{\eta}}$. For any positive sufficiently regular density q and vector field u,

$$\partial_t q + \nabla \cdot (q u) = q\left[(\partial_t + u \cdot \nabla) \log q + \nabla \cdot u\right]. \tag{69}$$

The reference continuity equation implies

$$(\partial_t + u_t \cdot \nabla) \log \widetilde{q}_t + \nabla \cdot u_t = 0. \tag{70}$$

Using equation 56, the time independence of ϕ, and $m_t = J_\phi u_t$,

$$(\partial_t + u_t \cdot \nabla) \log q_t + \nabla \cdot u_t = \dot{\lambda}_t^\top \phi + \lambda_t^\top m_t - \frac{d}{dt} A_t(\lambda_t). \tag{71}$$

Equations equation 58 and equation 63 give

$$\frac{d}{dt} A_t(\lambda_t) = \dot{\lambda}_t^\top c_t + \mathbb{E}_{Q_t^{\boldsymbol{\eta}}}[\lambda_t^\top m_t]. \tag{72}$$

Hence the bracket in equation 69 equals

$$\dot{\lambda}_t^\top (\phi - c_t) + \lambda_t^\top \left(m_t - \mathbb{E}_{Q_t^{\boldsymbol{\eta}}}[m_t]\right) = h_{t,\boldsymbol{\eta}}. \tag{73}$$

Substituting into equation 69 proves equation 17 and the explicit formula equation 18.

Finally, calibration gives $\mathbb{E}_{Q_t^{\boldsymbol{\eta}}}[\phi - c_t] = 0$, and the second centered term also has zero expectation. Therefore

$$\mathbb{E}_{Q_t^{\boldsymbol{\eta}}}[h_{t,\boldsymbol{\eta}}] = 0, \tag{74}$$

which is equation 19. Equivalently, the same identity follows by integrating equation 17 over space and using conservation of total mass. □

Corollary 3 (Density-free unbalanced forcing)**.** *Under the assumptions of Corollary 2, the normalized-shape residual satisfies*

$$h_{\mathrm{shape}}(x) = \dot{\lambda}_t^\top (\phi(x) - c_t) + \lambda_t^\top (m_t(x) - \mathbb{E}_{q_t}[m_t]), \tag{75}$$

and therefore the finite-measure reference-relative forcing in equation 40 is

$$h_{\mathrm{ub}}(x) = \dot{\lambda}_t^\top (\phi(x) - c_t) + \lambda_t^\top (m_t(x) - \mathbb{E}_{q_t}[m_t]) + \frac{\dot{M}_t}{M_t} - g_{\mathrm{ref}}(t). \tag{76}$$

Moreover,

$$\mathbb{E}_{q_t}[h_{\mathrm{shape}}] = 0, \qquad \mathbb{E}_{q_t}[h_{\mathrm{ub}}] = \frac{\dot{M}_t}{M_t} - g_{\mathrm{ref}}(t). \tag{77}$$

Proof. Equation equation 75 is Proposition 4 applied to the normalized shape. Adding the scalar mass-rate discrepancy from equation 40 gives equation 76. Taking expectation and using the centering in equation 75 gives equation 77. □

A.4 Minimum-energy full-law correction

For the weak formulation, let H_q denote the mean-zero weighted Sobolev energy space obtained by completing the admissible test functions satisfying $\mathbb{E}_q[\psi] = 0$ under

$$\|\psi\|_{H_q}^2 := \mathbb{E}_q[\|\nabla\psi\|^2]. \tag{78}$$

Assume that the weighted Dirichlet form is coercive on mean-zero functions, for example through a weighted Poincaré inequality, and that $\varphi \mapsto \mathbb{E}_q[h\varphi]$ defines a bounded linear functional on H_q. The weak Poisson equation is

$$\mathbb{E}_{Q_t^{\boldsymbol{\eta}}}[\nabla\psi_{t,\boldsymbol{\eta}} \cdot \nabla\varphi] = -\mathbb{E}_{Q_t^{\boldsymbol{\eta}}}[h_{t,\boldsymbol{\eta}}\varphi] \qquad \text{for every } \varphi \in H_q. \tag{79}$$

Proof of Theorem 2. Fix $(t, \boldsymbol{\eta})$ and use the shorthand $q = q_t^{\boldsymbol{\eta}}$ and $h = h_{t,\boldsymbol{\eta}}$. Define the bilinear form

$$a(\psi, \varphi) := \mathbb{E}_q[\nabla\psi \cdot \nabla\varphi] \tag{80}$$

on the mean-zero energy space H_q. By construction of the energy norm,

$$|a(\psi, \varphi)| \leq \|\psi\|_{H_q} \|\varphi\|_{H_q}, \qquad a(\psi, \psi) = \|\psi\|_{H_q}^2. \tag{81}$$

The weighted Poincaré/coercivity assumption ensures that the mean-zero gauge removes the constant null direction and that this energy space is the appropriate Hilbert space for the problem. By assumption,

$$\ell(\varphi) := -\mathbb{E}_q[h\varphi] \tag{82}$$

is a bounded linear functional on H_q. Lax–Milgram therefore gives a unique $\psi \in H_q$ satisfying

$$a(\psi, \varphi) = \ell(\varphi) \qquad \text{for every } \varphi \in H_q, \tag{83}$$

which is exactly equation 79. Since $\mathbb{E}_q[h] = 0$, the restriction to mean-zero test functions causes no loss: for any admissible φ, setting $\varphi_0 := \varphi - \mathbb{E}_q[\varphi]$ gives $\nabla\varphi_0 = \nabla\varphi$ and $\mathbb{E}_q[h\varphi_0] = \mathbb{E}_q[h\varphi]$, so the same weak identity holds for arbitrary admissible test functions. Under the stated periodic, decay, or no-flux boundary conditions, integration by parts converts this weak identity into

$$\nabla \cdot (q\nabla\psi) = qh \tag{84}$$

in the distributional sense, together with the gauge $\mathbb{E}_q[\psi] = 0$.

Set
$$\delta^* := -\nabla\psi. \tag{85}$$
Then $\delta^* \in L^2(q;\mathbb{R}^d)$. The weak Poisson equation gives, for every $\varphi \in H_q$,
$$\mathbb{E}_q[\delta^* \cdot \nabla\varphi] = \mathbb{E}_q[h\varphi]. \tag{86}$$
This is precisely the weak form of
$$\nabla \cdot (q\delta^*) = -qh, \tag{87}$$
so $u_t + \delta^*$ satisfies the projected-law continuity equation.

Now let $\delta \in L^2(q;\mathbb{R}^d)$ be any other feasible correction and define $w := \delta - \delta^*$. Subtracting the two weak constraints gives the identity first on the admissible test class. Because $w \in L^2(q;\mathbb{R}^d)$, the map $\varphi \mapsto \mathbb{E}_q[w \cdot \nabla\varphi]$ is continuous in the H_q energy norm; density therefore extends the identity to all of H_q:
$$\mathbb{E}_q[w \cdot \nabla\varphi] = 0 \qquad \text{for every } \varphi \in H_q. \tag{88}$$
Taking $\varphi = \psi$ yields
$$\mathbb{E}_q[w \cdot \nabla\psi] = 0, \tag{89}$$
and therefore, because $\delta^* = -\nabla\psi$,
$$\mathbb{E}_q[\delta^* \cdot w] = 0. \tag{90}$$
Consequently
$$\begin{aligned}\mathbb{E}_q[\|\delta\|^2] &= \mathbb{E}_q[\|\delta^* + w\|^2] \\ &= \mathbb{E}_q[\|\delta^*\|^2] + \mathbb{E}_q[\|w\|^2] \\ &\geq \mathbb{E}_q[\|\delta^*\|^2]. \end{aligned} \tag{91}$$
Equality holds if and only if $w = 0$ q-almost everywhere. Thus δ^* is the unique minimum-energy feasible correction, and
$$a_{\mathrm{full}}(t,\boldsymbol{\eta}) = \mathbb{E}_q[\|\delta^*\|^2] = \mathbb{E}_q[\|\nabla\psi\|^2], \tag{92}$$
proving equation 23. □

Corollary 4 (Minimum-energy unbalanced move–reaction correction)**.** *Fix $(t,\boldsymbol{\eta})$, write $q = q_t^{\boldsymbol{\eta}}$, $M = M_t$, and $h_{\mathrm{ub}} = h_{\mathrm{ub},t,\boldsymbol{\eta}}$, and let $\kappa > 0$. Let $H_{q,\kappa}$ be the completion of the admissible test functions under*
$$\|\psi\|^2_{H_{q,\kappa}} := \mathbb{E}_q\left[\|\nabla\psi\|^2 + \frac{\psi^2}{\kappa}\right]. \tag{93}$$
Assume that $\varphi \mapsto \mathbb{E}_q[h_{\mathrm{ub}}\varphi]$ is a bounded linear functional on $H_{q,\kappa}$. Then there exists a unique $\psi \in H_{q,\kappa}$ satisfying
$$\mathbb{E}_q\left[\nabla\psi \cdot \nabla\varphi + \frac{1}{\kappa}\psi\varphi\right] = \mathbb{E}_q[h_{\mathrm{ub}}\varphi] \qquad \textit{for every } \varphi \in H_{q,\kappa}. \tag{94}$$
Under the stated boundary and regularity assumptions, this is the weak form of
$$-\nabla\cdot(q\nabla\psi) + \frac{q}{\kappa}\psi = qh_{\mathrm{ub}}. \tag{95}$$
The unique minimum-energy correction in equation 42 is
$$\delta^* = \nabla\psi, \qquad \alpha^* = \frac{\psi}{\kappa}, \tag{96}$$
and
$$a^{\mathrm{ub}}_{\mathrm{full}}(t,\boldsymbol{\eta}) = M\mathbb{E}_q\left[\|\nabla\psi\|^2 + \frac{\psi^2}{\kappa}\right]. \tag{97}$$

Proof. Define
$$a_\kappa(\psi,\varphi) := \mathbb{E}_q\left[\nabla\psi \cdot \nabla\varphi + \frac{1}{\kappa}\psi\varphi\right]. \tag{98}$$

This bilinear form is continuous and coercive on $H_{q,\kappa}$ because $a_\kappa(\psi,\psi) = \|\psi\|^2_{H_{q,\kappa}}$. Lax–Milgram therefore gives a unique solution of equation 94. Integration by parts yields equation 95 whenever the strong form is justified.

Set $(\delta^*, \alpha^*) = (\nabla\psi, \psi/\kappa)$. Equation equation 94 gives

$$\mathbb{E}_q[\delta^* \cdot \nabla\varphi + \alpha^*\varphi] = \mathbb{E}_q[h_{\mathrm{ub}}\varphi], \tag{99}$$

which is the weak form of equation 41, so (δ^*, α^*) is feasible. Let (δ, α) be any other feasible pair and write $w = \delta - \delta^*$ and $\gamma = \alpha - \alpha^*$. Subtracting the two weak constraints gives

$$\mathbb{E}_q[w \cdot \nabla\varphi + \gamma\varphi] = 0 \tag{100}$$

for every $\varphi \in H_{q,\kappa}$. Taking $\varphi = \psi$ yields

$$\mathbb{E}_q[w \cdot \delta^* + \kappa\gamma\alpha^*] = \mathbb{E}_q[w \cdot \nabla\psi + \gamma\psi] = 0. \tag{101}$$

Hence, in the product norm induced by $\mathbb{E}_q[\|\delta\|^2 + \kappa\alpha^2]$,

$$\begin{aligned}\mathbb{E}_q[\|\delta\|^2 + \kappa\alpha^2] &= \mathbb{E}_q[\|\delta^*\|^2 + \kappa(\alpha^*)^2] + \mathbb{E}_q[\|w\|^2 + \kappa\gamma^2] \\ &\geq \mathbb{E}_q[\|\delta^*\|^2 + \kappa(\alpha^*)^2]. \end{aligned} \tag{102}$$

Equality holds only when $w = 0$ and $\gamma = 0$ in the corresponding $L^2(q)$ spaces. Multiplication by $M > 0$ proves both uniqueness and equation 97. ☐

Remark 2 (Screening removes the balanced compatibility and gauge conditions)**.** *The reaction term changes the functional analysis in a useful way. In the balanced problem, constants are a null direction of the Dirichlet seminorm, so one imposes a mean-zero gauge and requires $\mathbb{E}_q[h] = 0$. For finite κ, the term $\kappa^{-1}\mathbb{E}_q[\psi^2]$ controls constants directly, and neither a gauge condition nor zero-mean forcing is required. Indeed, testing equation 94 with $\varphi = 1$ gives $\mathbb{E}_q[\psi]/\kappa = \mathbb{E}_q[h_{\mathrm{ub}}]$, so the mean forcing is realized by the mean reaction rate.*

A.5 Weighted negative-Sobolev interpretation

Define the weighted negative-Sobolev norm

$$\|h\|_{H^{-1}(q)} := \sup_{\varphi \in H_q : \mathbb{E}_q[\|\nabla\varphi\|^2] \leq 1} |\mathbb{E}_q[h\varphi]| \,. \tag{103}$$

Corollary 5 (Weighted negative-Sobolev interpretation)**.** *Under the conditions of Theorem 2,*

$$a_{\mathrm{full}}(t, \boldsymbol{\eta}) = \|h_{t,\boldsymbol{\eta}}\|^2_{H^{-1}(q_t^{\boldsymbol{\eta}})} \,. \tag{104}$$

Proof of Corollary 5. On H_q use the inner product

$$\langle \psi, \varphi \rangle_{H_q} := \mathbb{E}_q[\nabla\psi \cdot \nabla\varphi]. \tag{105}$$

Theorem 2 gives

$$\mathbb{E}_q[h\varphi] = -\langle \psi, \varphi \rangle_{H_q} \tag{106}$$

for every $\varphi \in H_q$. Hence, by Cauchy–Schwarz,

$$|\mathbb{E}_q[h\varphi]| \leq \|\psi\|_{H_q} \|\varphi\|_{H_q} \,, \tag{107}$$

so $\|h\|_{H^{-1}(q)} \leq \|\psi\|_{H_q}$. If $\psi \neq 0$, choosing $\varphi = -\psi / \|\psi\|_{H_q}$ attains equality; if $\psi = 0$, both sides vanish. Therefore

$$\|h\|_{H^{-1}(q)} = \|\psi\|_{H_q} = \left(\mathbb{E}_q[\|\nabla\psi\|^2]\right)^{1/2} . \tag{108}$$

Squaring and applying equation 23 proves equation 104. ☐

Corollary 6 (Screened negative-Sobolev interpretation)**.** *Define*

$$\|h\|_{H_\kappa^{-1}(q)} := \sup_{\varphi \in H_{q,\kappa} : \mathbb{E}_q[\|\nabla\varphi\|^2 + \kappa^{-1}\varphi^2] \leq 1} |\mathbb{E}_q[h\varphi]| \,. \tag{109}$$

Under the conditions of Corollary 4,

$$a^{\mathrm{ub}}_{\mathrm{full}}(t, \boldsymbol{\eta}) = M_t \|h_{\mathrm{ub},t,\boldsymbol{\eta}}\|^2_{H_\kappa^{-1}(q_t^{\boldsymbol{\eta}})} \,. \tag{110}$$

Proof. Equation equation 94 states that the functional $\varphi \mapsto \mathbb{E}_q[h_{\mathrm{ub}}\varphi]$ is represented in the Hilbert space $H_{q,\kappa}$ by $\psi^\star$. Cauchy–Schwarz gives an upper bound by $\|\psi^\star\|_{H_{q,\kappa}}$, and choosing $\varphi = \psi^\star / \|\psi^\star\|_{H_{q,\kappa}}$ when $\psi^\star \neq 0$ attains equality. Corollary 4 then gives equation 110. ☐

A.6 MINIMUM-ENERGY TANGENT CORRECTION

For later reference, exact feasibility is

$$r_{t,\boldsymbol{\eta}} \in \mathrm{Range}(G_{t,\boldsymbol{\eta}}). \tag{111}$$

Proof of Proposition 1. Fix $(t, \boldsymbol{\eta})$ and let

$$\mathcal{H} := L^2(Q_t^{\boldsymbol{\eta}}; \mathbb{R}^d) \tag{112}$$

with its usual inner product. Define the bounded linear operator

$$L : \mathcal{H} \to \mathbb{R}^R, \qquad L\delta := \mathbb{E}_{Q_t^{\boldsymbol{\eta}}}[J_{\boldsymbol{\Phi}_{\boldsymbol{\eta}}}\delta]. \tag{113}$$

Boundedness follows from Cauchy–Schwarz and the assumption $J_{\boldsymbol{\Phi}_{\boldsymbol{\eta}}} \in L^2(Q_t^{\boldsymbol{\eta}})$. Its adjoint is

$$L^* a = J_{\boldsymbol{\Phi}_{\boldsymbol{\eta}}}^\top a, \qquad a \in \mathbb{R}^R, \tag{114}$$

because

$$\langle L\delta, a\rangle_{\mathbb{R}^R} = \mathbb{E}_{Q_t^{\boldsymbol{\eta}}}\left[\delta^\top J_{\boldsymbol{\Phi}_{\boldsymbol{\eta}}}^\top a\right]. \tag{115}$$

Moreover,

$$LL^* a = G_{t,\boldsymbol{\eta}} a. \tag{116}$$

Since the codomain is finite dimensional,

$$\mathrm{Range}(L) = \mathrm{Range}(LL^*) = \mathrm{Range}(G_{t,\boldsymbol{\eta}}). \tag{117}$$

Thus the condition $r_{t,\boldsymbol{\eta}} \in \mathrm{Range}(G_{t,\boldsymbol{\eta}})$ is exactly the feasibility condition for $L\delta = -r_{t,\boldsymbol{\eta}}$.

The feasible set is a closed affine subspace of the Hilbert space $\mathcal{H}$, so it contains a unique element of minimum norm. This minimum-norm element is orthogonal to $\ker L$ and therefore belongs to

$$(\ker L)^\perp = \overline{\mathrm{Range}(L^*)}. \tag{118}$$

Because L^* has finite-dimensional domain, its range is closed, so the minimizer has the form $\delta = L^* a$. The constraint becomes

$$G_{t,\boldsymbol{\eta}} a = -r_{t,\boldsymbol{\eta}}. \tag{119}$$

The minimum-norm coefficient may be chosen as

$$a = -G_{t,\boldsymbol{\eta}}^\dagger r_{t,\boldsymbol{\eta}}, \tag{120}$$

which gives

$$\delta_{t,\boldsymbol{\eta}}^{\tan}(x) = -J_{\boldsymbol{\Phi}_{\boldsymbol{\eta}}}(x)^\top G_{t,\boldsymbol{\eta}}^\dagger r_{t,\boldsymbol{\eta}}. \tag{121}$$

Its squared norm is

$$\begin{aligned}\mathbb{E}_{Q_t^{\boldsymbol{\eta}}}[\|\delta_{t,\boldsymbol{\eta}}^{\tan}\|^2] &= (G_{t,\boldsymbol{\eta}}^\dagger r_{t,\boldsymbol{\eta}})^\top G_{t,\boldsymbol{\eta}} (G_{t,\boldsymbol{\eta}}^\dagger r_{t,\boldsymbol{\eta}}) \\ &= r_{t,\boldsymbol{\eta}}^\top G_{t,\boldsymbol{\eta}}^\dagger r_{t,\boldsymbol{\eta}},\end{aligned} \tag{122}$$

where the last equality uses symmetry of $G_{t,\boldsymbol{\eta}}$ and $r_{t,\boldsymbol{\eta}} \in \mathrm{Range}(G_{t,\boldsymbol{\eta}})$. This proves equation 27–equation 28.

If $r_{t,\boldsymbol{\eta}} \notin \mathrm{Range}(G_{t,\boldsymbol{\eta}}) = \mathrm{Range}(L)$, the exact affine constraint is infeasible. In that case the pseudoinverse projects the requested rate onto the supported range, which is a least-squares statement rather than exact moment-rate matching. □

Corollary 7 (Minimum-energy unbalanced tangent correction)**.** *Let* $\bar{c}_t = \int \phi \, d\mu_t$ *denote the raw finite-measure moment vector, and assume* $\phi \in L^2(q_t; \mathbb{R}^R)$ *and* $J_\phi \in L^2(q_t; \mathbb{R}^{R\times d})$*. Define*

$$r_{t,\boldsymbol{\eta}}^{\mathrm{ub}} := \mathbb{E}_{q_t}[J_\phi u_t + g_{\mathrm{ref}}\phi] - \frac{\dot{\bar{c}}_t}{M_t} \tag{123}$$

and

$$G_{t,\boldsymbol{\eta}}^{\mathrm{ub}} := \mathbb{E}_{q_t}\left[J_\phi J_\phi^\top + \frac{1}{\kappa}\phi\phi^\top\right]. \tag{124}$$

If $r^{\rm ub}_{t,\boldsymbol{\eta}} \in \mathrm{Range}(G^{\rm ub}_{t,\boldsymbol{\eta}})$*, then the minimum of*

$$M_t \mathbb{E}_{q_t}[\|\delta\|^2 + \kappa\alpha^2] \tag{125}$$

subject to exact raw-moment-rate matching

$$\mathbb{E}_{q_t}[J_\phi \delta + \phi\alpha] = -r^{\rm ub}_{t,\boldsymbol{\eta}} \tag{126}$$

is attained uniquely by

$$\delta^{\rm tan}_{\rm ub} = -J_\phi^\top (G^{\rm ub})^\dagger r^{\rm ub}, \qquad \alpha^{\rm tan}_{\rm ub} = -\frac{1}{\kappa}\phi^\top (G^{\rm ub})^\dagger r^{\rm ub}, \tag{127}$$

and

$$a^{\rm ub}_{\tan}(t,\boldsymbol{\eta}) = M_t (r^{\rm ub}_{t,\boldsymbol{\eta}})^\top (G^{\rm ub}_{t,\boldsymbol{\eta}})^\dagger r^{\rm ub}_{t,\boldsymbol{\eta}}. \tag{128}$$

If $r^{\rm ub} \notin \mathrm{Range}(G^{\rm ub})$*, exact raw-moment-rate matching is infeasible.*

Proof. For any sufficiently regular finite-measure path satisfying

$$\partial_t \mu + \nabla\cdot(\mu v) = \mu g, \tag{129}$$

integration by parts gives

$$\frac{d}{dt}\int \phi \, d\mu = \int (J_\phi v + \phi g)\, d\mu. \tag{130}$$

Applying this with $v = u + \delta$, $g = g_{\rm ref} + \alpha$, and $\mu = Mq$ yields equation 126. Now equip

$$\mathcal{Z} := L^2(q;\mathbb{R}^d) \times L^2(q) \tag{131}$$

with

$$\langle(\delta,\alpha),(\delta',\alpha')\rangle_{\mathcal{Z}} := \mathbb{E}_q[\delta\cdot\delta' + \kappa\alpha\alpha']. \tag{132}$$

Define $L_{\rm ub} : \mathcal{Z} \to \mathbb{R}^R$ by $L_{\rm ub}(\delta,\alpha) = \mathbb{E}_q[J_\phi\delta + \phi\alpha]$. Its adjoint is

$$L^*_{\rm ub} a = \left(J_\phi^\top a, \frac{1}{\kappa}\phi^\top a\right), \tag{133}$$

since

$$\langle(\delta,\alpha), L^*_{\rm ub}a\rangle_{\mathcal{Z}} = a^\top \mathbb{E}_q[J_\phi\delta + \phi\alpha]. \tag{134}$$

Consequently,

$$L_{\rm ub}L^*_{\rm ub} a = G^{\rm ub} a. \tag{135}$$

The Hilbert-space minimum-norm argument from Proposition 1 therefore applies verbatim, giving $(\delta^{\rm tan}_{\rm ub}, \alpha^{\rm tan}_{\rm ub}) = -L^*_{\rm ub}(G^{\rm ub})^\dagger r^{\rm ub}$ and squared product norm $(r^{\rm ub})^\top (G^{\rm ub})^\dagger r^{\rm ub}$. Multiplication by M_t gives equation 128. □

Remark 3 (Making total-mass dynamics tangent-visible)**.** *If the raw tangent observable vector is augmented by the constant coordinate* $\phi_0 \equiv 1$*, then its transport Jacobian vanishes and its tangent equation reduces to the mass-rate condition. The corresponding Gram entry is* κ^{-1}*, so total-mass mismatch is represented entirely through reaction. This augmentation is appropriate for the tangent criterion even though, by Remark 1, the constant coordinate should not be inserted into the normalized-shape information-projection covariance.*

A.7 FULL–TANGENT–HIDDEN DECOMPOSITION

The pointwise identities underlying the balanced decomposition are

$$\mathbb{E}_{Q_t^{\boldsymbol{\eta}}}\left[J_{\boldsymbol{\Phi}_{\boldsymbol{\eta}}} \delta^{\rm hid}_{t,\boldsymbol{\eta}}\right] = 0, \tag{136}$$

and

$$\mathbb{E}_{Q_t^{\boldsymbol{\eta}}}\left[\delta^{\rm tan}_{t,\boldsymbol{\eta}} \cdot \delta^{\rm hid}_{t,\boldsymbol{\eta}}\right] = 0. \tag{137}$$

The hidden action density is

$$a_{\rm hid}(t,\boldsymbol{\eta}) := \mathbb{E}_{Q_t^{\boldsymbol{\eta}}}\left[\left\|\delta^{\rm hid}_{t,\boldsymbol{\eta}}\right\|^2\right] \geq 0. \tag{138}$$

Proof of Proposition 2. Fix $(t,\boldsymbol{\eta})$ and write $Q = Q_t^{\boldsymbol{\eta}}$. Let

$$L\delta := \mathbb{E}_Q[J_{\boldsymbol{\Phi}_{\boldsymbol{\eta}}}\delta] \tag{139}$$

be the moment-rate operator introduced implicitly in the proof of Proposition 1. Let δ_t^* denote the minimum-energy full-law correction. Because $u_t + \delta_t^*$ transports the projected law and the calibration identity gives $\mathbb{E}_Q[\boldsymbol{\Phi}_{\boldsymbol{\eta}}] = \boldsymbol{c}_{\boldsymbol{\eta}}(t)$, the moment-rate identity equation 24 yields

$$L\delta_t^* = -\left(\mathbb{E}_Q[J_{\boldsymbol{\Phi}_{\boldsymbol{\eta}}}u_t] - \dot{\boldsymbol{c}}_{\boldsymbol{\eta}}(t)\right) = -r_{t,\boldsymbol{\eta}}. \tag{140}$$

Hence the tangent constraint is feasible. By Proposition 1, its unique minimum-norm solution is $\delta_{t,\boldsymbol{\eta}}^{\tan}$.

Define

$$\delta_{t,\boldsymbol{\eta}}^{\mathrm{hid}} := \delta_t^* - \delta_{t,\boldsymbol{\eta}}^{\tan}. \tag{141}$$

Both δ_t^* and $\delta_{t,\boldsymbol{\eta}}^{\tan}$ satisfy $L\delta = -r_{t,\boldsymbol{\eta}}$, so

$$L\delta_{t,\boldsymbol{\eta}}^{\mathrm{hid}} = 0. \tag{142}$$

Thus $\delta_{t,\boldsymbol{\eta}}^{\mathrm{hid}} \in \ker L$, which proves equation 136.

The proof of Proposition 1 also shows that the minimum-norm tangent solution belongs to $(\ker L)^\perp$. Therefore

$$\mathbb{E}_Q\left[\delta_{t,\boldsymbol{\eta}}^{\tan} \cdot \delta_{t,\boldsymbol{\eta}}^{\mathrm{hid}}\right] = 0, \tag{143}$$

which is equation 137. Applying the Pythagorean identity in $L^2(Q;\mathbb{R}^d)$ gives

$$\begin{aligned}
a_{\mathrm{full}}(t,\boldsymbol{\eta}) &= \mathbb{E}_Q[\|\delta_t^*\|^2] \\
&= \mathbb{E}_Q\left[\left\|\delta_{t,\boldsymbol{\eta}}^{\tan} + \delta_{t,\boldsymbol{\eta}}^{\mathrm{hid}}\right\|^2\right] \\
&= \mathbb{E}_Q[\left\|\delta_{t,\boldsymbol{\eta}}^{\tan}\right\|^2] + \mathbb{E}_Q[\left\|\delta_{t,\boldsymbol{\eta}}^{\mathrm{hid}}\right\|^2] \\
&= a_{\tan}(t,\boldsymbol{\eta}) + a_{\mathrm{hid}}(t,\boldsymbol{\eta}),
\end{aligned} \tag{144}$$

proving equation 29. Integrating the pointwise identity against the common nonnegative measure ρ gives equation 30 whenever the terms are measurable. Nonnegativity of the squared norm yields the lower bounds in equation 31. At fixed t, equality $a_{\tan}(t,\boldsymbol{\eta}) = a_{\mathrm{full}}(t,\boldsymbol{\eta})$ holds if and only if $\delta_{t,\boldsymbol{\eta}}^{\mathrm{hid}} = 0$ in $L^2(Q;\mathbb{R}^d)$, equivalently $\delta_t^* = \delta_{t,\boldsymbol{\eta}}^{\tan}$ Q-almost everywhere. For the integrated actions, $\mathcal{A}_{\tan}(\boldsymbol{\eta}) = \mathcal{A}(\boldsymbol{\eta})$ holds if and only if $a_{\mathrm{hid}}(t,\boldsymbol{\eta}) = 0$ for ρ-almost every t, equivalently $\delta_{t,\boldsymbol{\eta}}^{\mathrm{hid}} = 0$ in $L^2(Q_t^{\boldsymbol{\eta}};\mathbb{R}^d)$ for ρ-almost every t. $\square$

Corollary 8 (Unbalanced Full–Tangent–Hidden decomposition)**.** *Under the assumptions of Corollaries 4 and 7, let*

$$z_{\mathrm{ub}}^* := (\delta^*,\alpha^*), \qquad z_{\mathrm{ub}}^{\tan} := (\delta_{\mathrm{ub}}^{\tan},\alpha_{\mathrm{ub}}^{\tan}), \qquad z_{\mathrm{ub}}^{\mathrm{hid}} := z_{\mathrm{ub}}^* - z_{\mathrm{ub}}^{\tan}. \tag{145}$$

Then

$$L_{\mathrm{ub}}z_{\mathrm{ub}}^{\mathrm{hid}} = 0 \tag{146}$$

and

$$\mathbb{E}_q\left[\delta_{\mathrm{ub}}^{\tan} \cdot \delta_{\mathrm{ub}}^{\mathrm{hid}} + \kappa\alpha_{\mathrm{ub}}^{\tan}\alpha_{\mathrm{ub}}^{\mathrm{hid}}\right] = 0. \tag{147}$$

Defining

$$a_{\mathrm{hid}}^{\mathrm{ub}}(t,\boldsymbol{\eta}) := M_t\mathbb{E}_q\left[\|\delta_{\mathrm{ub}}^{\mathrm{hid}}\|^2 + \kappa(\alpha_{\mathrm{ub}}^{\mathrm{hid}})^2\right], \tag{148}$$

one has

$$a_{\mathrm{full}}^{\mathrm{ub}}(t,\boldsymbol{\eta}) = a_{\tan}^{\mathrm{ub}}(t,\boldsymbol{\eta}) + a_{\mathrm{hid}}^{\mathrm{ub}}(t,\boldsymbol{\eta}), \qquad a_{\tan}^{\mathrm{ub}}(t,\boldsymbol{\eta}) \le a_{\mathrm{full}}^{\mathrm{ub}}(t,\boldsymbol{\eta}). \tag{149}$$

Whenever the three terms are integrable against the common nonnegative measure ρ, the same decomposition and inequality hold after integration in time.

Proof. The Full correction satisfies the complete finite-measure continuity equation, so the raw moment identity equation 130 implies that z_{ub}^* satisfies the tangent constraint. The tangent correction is the unique minimum-norm point of that affine constraint in the product Hilbert space equation 132; therefore $z_{\mathrm{ub}}^{\tan} \in (\ker L_{\mathrm{ub}})^\perp$. Since both z_{ub}^* and $z_{\mathrm{ub}}^{\tan}$ have the same image under L_{ub}, their difference lies in $\ker L_{\mathrm{ub}}$. This proves equation 146 and equation 147. Pythagoras in the product Hilbert space, followed by multiplication by M_t, gives equation 149; integration against ρ preserves the identity and inequality. $\square$

When $\mathcal{A}(\boldsymbol{\eta}) > 0$, we summarize the relative size of the tangent-invisible component by the hidden-action fraction

$$\Gamma(\boldsymbol{\eta}) := \frac{\mathcal{A}_{\mathrm{hid}}(\boldsymbol{\eta})}{\mathcal{A}(\boldsymbol{\eta})} = 1 - \frac{\mathcal{A}_{\tan}(\boldsymbol{\eta})}{\mathcal{A}(\boldsymbol{\eta})} \in [0, 1]. \tag{150}$$

Thus $\Gamma(\boldsymbol{\eta})$ near one indicates that most of the minimum law-level correction lies outside the directions visible through the measured moment rates. In the discretized experimental audits we denote the corresponding common-raster quantity by Γ_h.

A.8 Exact condition for tangent–full equality

Corollary 9 (When tangent and full transportability coincide)**.** *Under the conditions of Proposition 2, suppose additionally that $Q_t^{\boldsymbol{\eta}}$ has a density positive almost everywhere on a connected open domain Ω, and that $\psi_{t,\boldsymbol{\eta}}$ and the coordinates of $\boldsymbol{\Phi}_{\boldsymbol{\eta}}$ belong to $H^1_{\mathrm{loc}}(\Omega)$. Define*

$$\beta_{t,\boldsymbol{\eta}} := G^{\dagger}_{t,\boldsymbol{\eta}} r_{t,\boldsymbol{\eta}}. \tag{151}$$

Then

$$a_{\tan}(t, \boldsymbol{\eta}) = a_{\mathrm{full}}(t, \boldsymbol{\eta}) \tag{152}$$

if and only if

$$\psi_{t,\boldsymbol{\eta}}(x) = \beta^{\top}_{t,\boldsymbol{\eta}}(\boldsymbol{\Phi}_{\boldsymbol{\eta}}(x) - \boldsymbol{c}_{\boldsymbol{\eta}}(t)) \qquad Q_t^{\boldsymbol{\eta}}\text{-a.e.} \tag{153}$$

Proof of Corollary 9. Let

$$\beta_{t,\boldsymbol{\eta}} = G^{\dagger}_{t,\boldsymbol{\eta}} r_{t,\boldsymbol{\eta}}. \tag{154}$$

By Proposition 1,

$$\delta^{\tan}_{t,\boldsymbol{\eta}} = -J^{\top}_{\boldsymbol{\Phi}_{\boldsymbol{\eta}}} \beta_{t,\boldsymbol{\eta}} = -\nabla\left(\beta^{\top}_{t,\boldsymbol{\eta}} \boldsymbol{\Phi}_{\boldsymbol{\eta}}\right) \tag{155}$$

where the final identity is understood in the weak sense. Theorem 2 gives

$$\delta^*_t = -\nabla \psi_{t,\boldsymbol{\eta}}. \tag{156}$$

By Proposition 2, equality of tangent and full action is equivalent to $\delta^*_t = \delta^{\tan}_{t,\boldsymbol{\eta}}$ $Q_t^{\boldsymbol{\eta}}$-almost everywhere. Because the density of $Q_t^{\boldsymbol{\eta}}$ is positive almost everywhere on Ω, this is equivalent to

$$\nabla\left(\psi_{t,\boldsymbol{\eta}} - \beta^{\top}_{t,\boldsymbol{\eta}} \boldsymbol{\Phi}_{\boldsymbol{\eta}}\right) = 0 \tag{157}$$

almost everywhere on Ω. A Sobolev function with zero weak gradient on a connected open domain is almost everywhere constant. Hence

$$\psi_{t,\boldsymbol{\eta}}(x) = \beta^{\top}_{t,\boldsymbol{\eta}} \boldsymbol{\Phi}_{\boldsymbol{\eta}}(x) + C \tag{158}$$

for some constant C. The mean-zero gauge for the Poisson potential and the calibration identity imply

$$0 = \mathbb{E}_{Q_t^{\boldsymbol{\eta}}}[\psi_{t,\boldsymbol{\eta}}] = \beta^{\top}_{t,\boldsymbol{\eta}} \boldsymbol{c}_{\boldsymbol{\eta}}(t) + C, \tag{159}$$

so $C = -\beta^{\top}_{t,\boldsymbol{\eta}} \boldsymbol{c}_{\boldsymbol{\eta}}(t)$, giving equation 153.

Conversely, if equation 153 holds, then

$$-\nabla \psi_{t,\boldsymbol{\eta}} = -J^{\top}_{\boldsymbol{\Phi}_{\boldsymbol{\eta}}} \beta_{t,\boldsymbol{\eta}} = \delta^{\tan}_{t,\boldsymbol{\eta}} \tag{160}$$

$Q_t^{\boldsymbol{\eta}}$-almost everywhere. Thus the hidden component vanishes, and Proposition 2 gives $a_{\mathrm{full}}(t, \boldsymbol{\eta}) = a_{\tan}(t, \boldsymbol{\eta})$. ☐

Corollary 10 (Exact unbalanced tangent–Full equality condition)**.** *Under the assumptions of Corollary 8, assume that the screened-Poisson potential ψ and the coordinates of ϕ belong to $H_{q,\kappa}$, and define*

$$\beta^{\mathrm{ub}} := (G^{\mathrm{ub}})^{\dagger} r^{\mathrm{ub}}. \tag{161}$$

Then

$$a^{\mathrm{ub}}_{\tan}(t, \boldsymbol{\eta}) = a^{\mathrm{ub}}_{\mathrm{full}}(t, \boldsymbol{\eta}) \tag{162}$$

if and only if

$$\psi = -(\beta^{\mathrm{ub}})^{\top} \phi \qquad \text{in } H_{q,\kappa}, \tag{163}$$

equivalently, the screened Full potential belongs to the uncentered linear span of the tangent observables with the coefficient fixed by the tangent normal equations.

Proof. Corollary 8 gives equality if and only if the Full and Tangent correction pairs coincide. By Corollaries 4 and 7,

$$z^*_{\mathrm{ub}} = \left(\nabla\psi, \frac{\psi}{\kappa}\right), \qquad z^{\tan}_{\mathrm{ub}} = \left(-J_\phi^\top \beta^{\mathrm{ub}}, -\frac{1}{\kappa}\phi^\top \beta^{\mathrm{ub}}\right). \tag{164}$$

Equality of the reaction components gives $\psi = -(\beta^{\mathrm{ub}})^\top \phi$ in $L^2(q)$; because both sides lie in $H_{q,\kappa}$, the corresponding weak gradients then agree as well, giving equality of the transport components. Conversely, equation 163 implies equality of both components and hence zero hidden action. Unlike the balanced case, no additive constant remains undetermined because the reaction component contains the potential itself. □

A.9 No universal tangent control

The construction below proves the stronger zero-versus-positive statement

$$a_{\tan}(t, \boldsymbol{\eta}) = 0 \qquad \text{while} \qquad a_{\mathrm{full}}(t, \boldsymbol{\eta}) > 0 \tag{165}$$

for a smooth regular instance. The same construction gives a regular one-parameter family with

$$a_{\tan}(t, \boldsymbol{\eta}) \to 0, \qquad a_{\mathrm{full}}(t, \boldsymbol{\eta}) \to c > 0, \tag{166}$$

and therefore the divergence in equation 33.

Proof of Theorem 3. Let the state space be the one-dimensional torus $\Omega = \mathbb{T} = \mathbb{R}/(2\pi\mathbb{Z})$ and take the scalar observable

$$\Phi(x) = \cos x. \tag{167}$$

Fix constants $\lambda \neq 0$ and $a > 0$, and define

$$I_0(z) := \frac{1}{2\pi}\int_0^{2\pi} e^{z\cos x}\,dx, \qquad \widetilde{q}(x) := \frac{e^{-\lambda\cos x}}{2\pi I_0(\lambda)}, \qquad u(x) := ae^{\lambda\cos x}. \tag{168}$$

These functions are smooth, $\widetilde{q}$ is strictly positive, and u is bounded because $\mathbb{T}$ is compact. Moreover,

$$\widetilde{q}(x)u(x) = \frac{a}{2\pi I_0(\lambda)} \tag{169}$$

is spatially constant, so the time-independent pair $(\widetilde{q}, u)$ satisfies the reference continuity equation

$$\partial_t\widetilde{q} + \partial_x(\widetilde{q}u) = 0. \tag{170}$$

Fix a time t_0 and a scalar parameter $\varepsilon \in \mathbb{R}$. On a sufficiently small interval about t_0, define

$$\lambda_t := \lambda + 2\varepsilon(t - t_0) \tag{171}$$

and the exponentially tilted law

$$q_t(x) := \widetilde{q}(x)\exp(\lambda_t\Phi(x) - A(\lambda_t)), \qquad A(\alpha) := \log \mathbb{E}_{\widetilde{Q}}\left[e^{\alpha\Phi(X)}\right]. \tag{172}$$

Let

$$c(t) := \mathbb{E}_{q_t}[\Phi]. \tag{173}$$

By Theorem 1, q_t is the information projection of $\widetilde{q}$ onto the moment fiber determined by $c(t)$; if desired, taking the scientific law $P_t = Q_t$ realizes this moment trajectory within the problem setup. At $t = t_0$, $\lambda_{t_0} = \lambda$, and

$$A(\lambda) = \log \mathbb{E}_{\widetilde{Q}}[e^{\lambda\cos X}] = -\log I_0(\lambda), \tag{174}$$

so

$$q_{t_0}(x) = \frac{e^{-\lambda\cos x}}{2\pi I_0(\lambda)} e^{\lambda\cos x + \log I_0(\lambda)} = \frac{1}{2\pi}. \tag{175}$$

Thus the projected law is uniform at the comparison time. In particular,

$$c(t_0) = 0, \qquad C_{t_0} = \mathrm{Var}_{q_{t_0}}(\cos X) = \frac{1}{2}, \tag{176}$$

so the observable covariance is nonsingular. Differentiating the exponential family gives

$$\dot{c}(t_0) = C_{t_0} \dot{\lambda}_{t_0} = \frac{1}{2}(2\varepsilon) = \varepsilon. \tag{177}$$

The observable response to the reference velocity is

$$m(x) = \Phi'(x)u(x) = -a \sin x \, e^{\lambda \cos x}. \tag{178}$$

Since q_{t_0} is uniform,

$$\mathbb{E}_{q_{t_0}}[m] = -\frac{a}{2\pi} \int_0^{2\pi} \sin x \, e^{\lambda \cos x} \, dx = 0. \tag{179}$$

Hence the tangent residual from equation 25 is

$$r_{t_0} = \mathbb{E}_{q_{t_0}}[m] - \dot{c}(t_0) = -\varepsilon. \tag{180}$$

Furthermore,

$$G_{t_0} = \mathbb{E}_{q_{t_0}}[(\Phi'(X))^2] = \mathbb{E}_{q_{t_0}}[\sin^2 X] = \frac{1}{2}. \tag{181}$$

Proposition 1 therefore yields

$$a_{\tan}(t_0) = r_{t_0}^2 G_{t_0}^{-1} = 2\varepsilon^2. \tag{182}$$

We now compute the full-law action. From equation 172,

$$\partial_t \log q_t(x) = \dot{\lambda}_t \left(\Phi(x) - \mathbb{E}_{q_t}[\Phi] \right), \tag{183}$$

and therefore, using equation 175–equation 177,

$$\left. \frac{\partial_t q_t(x)}{q_t(x)} \right|_{t=t_0} = 2\varepsilon \cos x. \tag{184}$$

Because q_{t_0} is spatially constant,

$$\frac{\partial_x (q_{t_0} u)}{q_{t_0}} = u'(x) = -a\lambda \sin x \, e^{\lambda \cos x}. \tag{185}$$

Thus the normalized continuity residual equation 16 is

$$h_{t_0}(x) = 2\varepsilon \cos x + u'(x). \tag{186}$$

Since $q_{t_0} = (2\pi)^{-1}$, the weighted Poisson equation of Theorem 2 reduces to

$$\psi''(x) = h_{t_0}(x) = 2\varepsilon \cos x + u'(x). \tag{187}$$

Hence

$$\psi'(x) = 2\varepsilon \sin x + u(x) - \mathbb{E}_{q_{t_0}}[u], \tag{188}$$

where the additive constant in ψ' is fixed by periodicity, equivalently by the zero-integral condition on ψ' over one period. Theorem 2 now gives

$$\begin{aligned} a_{\text{full}}(t_0) &= \mathbb{E}_{q_{t_0}}[(\psi'(X))^2] \\ &= 4\varepsilon^2 \mathbb{E}_{q_{t_0}}[\sin^2 X] + \operatorname{Var}_{q_{t_0}}(u) + 4\varepsilon \mathbb{E}_{q_{t_0}}\left[\sin X \big(u(X) - \mathbb{E}_{q_{t_0}}[u] \big)\right]. \end{aligned} \tag{189}$$

The final term vanishes because $u(x) = a e^{\lambda \cos x}$ is even under $x \mapsto 2\pi - x$ whereas $\sin x$ is odd. Therefore

$$a_{\text{full}}(t_0) = 2\varepsilon^2 + \operatorname{Var}_{q_{t_0}}(u). \tag{190}$$

By the definition of I_0,

$$\mathbb{E}_{q_{t_0}}[u] = a I_0(\lambda), \qquad \mathbb{E}_{q_{t_0}}[u^2] = a^2 I_0(2\lambda), \tag{191}$$

and hence

$$\operatorname{Var}_{q_{t_0}}(u) = a^2 \left[I_0(2\lambda) - I_0(\lambda)^2 \right]. \tag{192}$$

Because $\lambda \neq 0$, the function $e^{\lambda \cos x}$ is nonconstant, so the variance in equation 192 is strictly positive. Combining equation 182 and equation 190 gives

$$a_{\tan}(t_0) = 2\varepsilon^2, \qquad a_{\text{full}}(t_0) = 2\varepsilon^2 + a^2 \left[I_0(2\lambda) - I_0(\lambda)^2 \right]. \tag{193}$$

Setting $\varepsilon = 0$ yields

$$a_{\tan}(t_0) = 0, \qquad a_{\text{full}}(t_0) = a^2 \left[I_0(2\lambda) - I_0(\lambda)^2\right] > 0, \tag{194}$$

which already rules out any universal finite K in equation 32. For $\varepsilon \neq 0$,

$$\frac{a_{\text{full}}(t_0)}{a_{\tan}(t_0)} = 1 + \frac{a^2 \left[I_0(2\lambda) - I_0(\lambda)^2\right]}{2\varepsilon^2}, \tag{195}$$

so as $\varepsilon \to 0$,

$$a_{\tan}(t_0) \to 0, \qquad a_{\text{full}}(t_0) \to a^2 \left[I_0(2\lambda) - I_0(\lambda)^2\right] > 0, \qquad \frac{a_{\text{full}}(t_0)}{a_{\tan}(t_0)} \to \infty. \tag{196}$$

This proves equation 165–equation 33. □

Corollary 11 (No universal tangent control in the unbalanced geometry)**.** *Fix any $\kappa > 0$. There is no universal finite constant K such that*

$$a^{\text{ub}}_{\text{full}}(t, \boldsymbol{\eta}) \leq K\, a^{\text{ub}}_{\tan}(t, \boldsymbol{\eta}) \tag{197}$$

for all regular finite-measure instances of the unbalanced construction. In fact, there are smooth instances with

$$a^{\text{ub}}_{\tan}(t, \boldsymbol{\eta}) = 0 \qquad \textit{and} \qquad a^{\text{ub}}_{\text{full}}(t, \boldsymbol{\eta}) > 0. \tag{198}$$

Proof. Use the construction in the proof of Theorem 3, set $M_t \equiv 1$ and $g_{\text{ref}} \equiv 0$, and evaluate at $t = t_0$, where q_{t_0} is uniform and

$$h_{\text{ub}}(x) = 2\varepsilon \cos x + u'(x). \tag{199}$$

For the single observable $\Phi(x) = \cos x$, Corollary 7 gives

$$G^{\text{ub}} = \mathbb{E}[\sin^2 X] + \frac{1}{\kappa}\mathbb{E}[\cos^2 X] = \frac{1}{2}\left(1 + \frac{1}{\kappa}\right), \tag{200}$$

and therefore

$$a^{\text{ub}}_{\tan}(t_0) = \frac{2\kappa}{\kappa + 1}\varepsilon^2. \tag{201}$$

For the Full problem, the screened operator on the uniform torus is

$$\mathcal{L}_\kappa := -\partial_{xx} + \kappa^{-1}. \tag{202}$$

The forcing $2\varepsilon \cos x$ is even, whereas $u'(x)$ is odd, and $\mathcal{L}_\kappa^{-1}$ preserves parity. Hence the two forcing components are orthogonal in the screened energy geometry. Since

$$\mathcal{L}_\kappa(\cos x) = \left(1 + \frac{1}{\kappa}\right)\cos x, \tag{203}$$

the even component contributes exactly

$$\frac{2\kappa}{\kappa + 1}\varepsilon^2. \tag{204}$$

The odd component contributes

$$c_\kappa := \mathbb{E}_{q_{t_0}}\left[u'\, \mathcal{L}_\kappa^{-1} u'\right] > 0, \tag{205}$$

because $\mathcal{L}_\kappa$ is strictly positive and $u' \not\equiv 0$. Thus

$$a^{\text{ub}}_{\text{full}}(t_0) = \frac{2\kappa}{\kappa + 1}\varepsilon^2 + c_\kappa. \tag{206}$$

At $\varepsilon = 0$, equation 201–equation 206 give equation 198. For $\varepsilon \neq 0$ their ratio diverges as $\varepsilon \to 0$, ruling out any universal finite K. □

A.10 Differentiability with respect to the measurement design

Define the calibration residual

$$F(\boldsymbol{\lambda}, \boldsymbol{\eta}, t) := \mathbb{E}_{q_{\boldsymbol{\lambda},\boldsymbol{\eta},t}}[\boldsymbol{\Phi}_{\boldsymbol{\eta}}] - \boldsymbol{c}_{\boldsymbol{\eta}}(t), \tag{207}$$

where

$$q_{\boldsymbol{\lambda},\boldsymbol{\eta},t}(x) \propto \widetilde{q}_t(x) \exp\bigl(\boldsymbol{\lambda}^\top \boldsymbol{\Phi}_{\boldsymbol{\eta}}(x)\bigr) . \tag{208}$$

Proposition 5 (Design sensitivity of the information projection)**.** *Suppose $F(\boldsymbol{\lambda}_{t,\boldsymbol{\eta}}, \boldsymbol{\eta}, t) = 0$, the relevant derivatives exist, differentiation and integration may be interchanged, and $C_{t,\boldsymbol{\eta}} = \partial_{\boldsymbol{\lambda}} F$ is positive definite. Then, for every design perturbation ξ,*

$$D_{\boldsymbol{\eta}} \boldsymbol{\lambda}_{t,\boldsymbol{\eta}}[\xi] = -C_{t,\boldsymbol{\eta}}^{-1} D_{\boldsymbol{\eta}} F[\xi], \tag{209}$$

where

$$\begin{aligned} D_{\boldsymbol{\eta}} F[\xi] = {} & \mathbb{E}_{Q_t^{\boldsymbol{\eta}}}[D_{\boldsymbol{\eta}} \boldsymbol{\Phi}_{\boldsymbol{\eta}}[\xi]] + \mathrm{Cov}_{Q_t^{\boldsymbol{\eta}}}\bigl(\boldsymbol{\Phi}_{\boldsymbol{\eta}}, \boldsymbol{\lambda}_{t,\boldsymbol{\eta}}^\top D_{\boldsymbol{\eta}} \boldsymbol{\Phi}_{\boldsymbol{\eta}}[\xi]\bigr) \\ & - D_{\boldsymbol{\eta}} \boldsymbol{c}_{\boldsymbol{\eta}}(t)[\xi]. \end{aligned} \tag{210}$$

Proof of Proposition 5. Fix t and abbreviate

$$q_{\lambda,\boldsymbol{\eta}}(x) = \widetilde{q}_t(x) \exp\bigl(\lambda^\top \boldsymbol{\Phi}_{\boldsymbol{\eta}}(x) - A_t(\lambda, \boldsymbol{\eta})\bigr) , \tag{211}$$

where

$$A_t(\lambda, \boldsymbol{\eta}) = \log \mathbb{E}_{\widetilde{Q}_t}\left[\exp(\lambda^\top \boldsymbol{\Phi}_{\boldsymbol{\eta}}(X))\right] . \tag{212}$$

The reference law is frozen with respect to $\boldsymbol{\eta}$. At a calibrated solution,

$$F(\lambda_{t,\boldsymbol{\eta}}, \boldsymbol{\eta}, t) = 0. \tag{213}$$

Differentiation with respect to λ gives

$$\partial_\lambda F = \mathrm{Cov}_{q_{\lambda,\boldsymbol{\eta}}}(\boldsymbol{\Phi}_{\boldsymbol{\eta}}, \boldsymbol{\Phi}_{\boldsymbol{\eta}}), \tag{214}$$

so at the calibrated point

$$\partial_\lambda F = C_{t,\boldsymbol{\eta}}. \tag{215}$$

By assumption this matrix is positive definite and hence invertible. The implicit-function theorem therefore gives a locally differentiable multiplier map $\boldsymbol{\eta} \mapsto \lambda_{t,\boldsymbol{\eta}}$ and, for any perturbation direction ξ,

$$D_{\boldsymbol{\eta}} \lambda_{t,\boldsymbol{\eta}}[\xi] = -C_{t,\boldsymbol{\eta}}^{-1} D_{\boldsymbol{\eta}} F[\xi]. \tag{216}$$

It remains to compute the directional derivative of F while holding λ fixed.

For any scalar or vector test function $f_{\boldsymbol{\eta}}$, differentiation of a normalized exponential tilt gives the identity

$$D_{\boldsymbol{\eta}} \mathbb{E}_{q_{\lambda,\boldsymbol{\eta}}}[f_{\boldsymbol{\eta}}][\xi] = \mathbb{E}_{q_{\lambda,\boldsymbol{\eta}}}[D_{\boldsymbol{\eta}} f_{\boldsymbol{\eta}}[\xi]] + \mathrm{Cov}_{q_{\lambda,\boldsymbol{\eta}}}\bigl(f_{\boldsymbol{\eta}}, \lambda^\top D_{\boldsymbol{\eta}} \boldsymbol{\Phi}_{\boldsymbol{\eta}}[\xi]\bigr) , \tag{217}$$

provided differentiation and integration can be interchanged. Applying equation 217 with $f_{\boldsymbol{\eta}} = \boldsymbol{\Phi}_{\boldsymbol{\eta}}$ and subtracting $D_{\boldsymbol{\eta}} \boldsymbol{c}_{\boldsymbol{\eta}}(t)[\xi]$ gives

$$\begin{aligned} D_{\boldsymbol{\eta}} F[\xi] = {} & \mathbb{E}_{Q_t^{\boldsymbol{\eta}}}[D_{\boldsymbol{\eta}} \boldsymbol{\Phi}_{\boldsymbol{\eta}}[\xi]] + \mathrm{Cov}_{Q_t^{\boldsymbol{\eta}}}\bigl(\boldsymbol{\Phi}_{\boldsymbol{\eta}}, \boldsymbol{\lambda}_{t,\boldsymbol{\eta}}^\top D_{\boldsymbol{\eta}} \boldsymbol{\Phi}_{\boldsymbol{\eta}}[\xi]\bigr) \\ & - D_{\boldsymbol{\eta}} \boldsymbol{c}_{\boldsymbol{\eta}}(t)[\xi], \end{aligned} \tag{218}$$

which is equation 210. Substitution into equation 216 proves equation 209. □

Corollary 12 (Design sensitivity in the finite-measure unbalanced extension)**.** *Suppose the normalized-shape information projection in Corollary 1 is calibrated to*

$$c_{\boldsymbol{\eta}}(t) = \frac{\bar{c}_{\boldsymbol{\eta}}(t)}{M_{\boldsymbol{\eta}}(t)}, \tag{219}$$

and suppose the assumptions of Proposition 5 hold. Then

$$D_{\boldsymbol{\eta}} \lambda_{t,\boldsymbol{\eta}}[\xi] = -C_{t,\boldsymbol{\eta}}^{-1} D_{\boldsymbol{\eta}} F[\xi] \tag{220}$$

with the same expression equation 210, but with

$$D_{\boldsymbol{\eta}} c_{\boldsymbol{\eta}}(t)[\xi] = \frac{1}{M_{\boldsymbol{\eta}}(t)} D_{\boldsymbol{\eta}} \bar{c}_{\boldsymbol{\eta}}(t)[\xi] - \frac{c_{\boldsymbol{\eta}}(t)}{M_{\boldsymbol{\eta}}(t)} D_{\boldsymbol{\eta}} M_{\boldsymbol{\eta}}(t)[\xi]. \tag{221}$$

In particular, if the mass schedule is fixed with respect to the measurement design, only the first term in equation 221 remains.

B Experimental Details and Reproducibility

This appendix gives the experimental specification for the three protocols summarized in Section 6: the analytic Gaussian-mixture benchmark, the retrospective double-gyre benchmark, and the aggregate-only prospective double-gyre benchmark. All three instantiate the balanced probability-law construction of Sections 4–5: endpoint information is used to train a reference dynamics that is frozen during each design stage; aggregate sensor information is reconstructed into moment trajectories; the corresponding frozen reference is information-projected onto those moments; and scientific risk and Full action are evaluated using physical-density weighted-Poisson discretizations. The first two protocols use hidden benchmark-population information during selection and evaluate on disjoint frozen validation or holdout data. The prospective protocol enforces a stricter information boundary: selection sees only endpoints and aggregate predictive fields, all geometries are frozen, and only then are the independent E1 validation reference and hidden validation trials generated.

B.1 Common risk-controlled selection protocol

For each protocol, a numerically optimized Law geometry is selected and frozen, and its audited finite-data selection risk $R_{\text{anchor}} > 0$ defines the operational risk baseline. For allowance p, Full and any other action-based design considered in that protocol must satisfy

$$R(\boldsymbol{\eta}) \le \left(1 + \frac{p}{100}\right) R_{\text{anchor}}. \tag{222}$$

The analytic Gaussian-mixture and retrospective double-gyre benchmarks additionally impose a fixed population-level scientific screen

$$L(\boldsymbol{\eta}) \le L^\star + \epsilon_L. \tag{223}$$

because their hidden population laws are available to the benchmarker during selection. This extra screen is benchmark-specific and is not part of the general FIDE deployment definition. The prospective protocol does not expose intermediate hidden target states during selection; its scientific-risk calculation is constructed from the frozen aggregate predictive interface instead.

The analytic benchmark evaluates $p \in \{0.5, 1, 2, 3, 4, 5\}\%$, whereas both double-gyre protocols evaluate $p \in \{0.5, 1, 2\}\%$. The analytic experiment uses one frozen reference and a disjoint 128-trial validation bank. The retrospective double-gyre experiment evaluates candidates across three independently trained frozen references and confirms the frozen Law and Full geometries on a fresh shared 64-trial holdout. The prospective experiment uses one frozen D0 reference for selection; after the repaired Law and Full frontier is completely frozen, an independent E1 endpoint reference and 64 fresh paired hidden-validation trials are generated for evaluation. In all three protocols, evaluation information is excluded from geometry selection. Since the continuous design problems are solved numerically, every frozen Law geometry is an operational finite-search risk anchor rather than a certificate of the global risk optimum.

B.2 Analytic Gaussian-mixture transport benchmark

B.2.1 Hidden population

The state is $x = (x_1, x_2)$ on $\Omega = [-3.2, 3.2]^2$. For orientation α, define

$$g_\alpha(x) = \frac{1}{2}\mathcal{N}\big(x; r\, d(\alpha), \sigma^2 I\big) + \frac{1}{2}\mathcal{N}\big(x; -r\, d(\alpha), \sigma^2 I\big), \qquad d(\alpha) = (\cos\alpha, \sin\alpha), \tag{224}$$

with $r = 1.5$ and $\sigma = 0.3$. The hidden population path is

$$\rho_t^\alpha(x) = (1-t)^2 g_0(x) + 2t(1-t) g_\alpha(x) + t^2 g_{\pi/2}(x), \qquad t \in [0, 1]. \tag{225}$$

The nuisance orientation is averaged uniformly over $\alpha \in [30^\circ, 60^\circ]$ using five-point Gauss–Legendre quadrature. All nuisance settings therefore share identical horizontal and vertical endpoint laws while differing at intermediate times. Only the common endpoint distributions are available to the learned reference model.

For deterministic reference integration, the initial law is represented by tensor-product Gauss–Hermite quadrature of order 36 per spatial coordinate for each lobe, producing $2 \times 36^2 = 2592$ weighted reference particles.

Figure 3: Observation mechanism for the analytic Gaussian-mixture benchmark. Columns show representative time snapshots from a frozen validation trial at the authoritative 5% Full geometry. The upper row is the hidden analytic population, while the middle row is the endpoint-only reference after information projection onto the two reconstructed sensor moments. The lower panels visualize the localized spatial weighting associated with each scalar sensor observation. The projected law is constrained to reproduce these aggregate moments, not the hidden density pointwise; differences away from the sensor supports therefore illustrate the unresolved law-level directions whose dynamical cost is measured by Full action.

B.2.2 Measurement model

A design is $\boldsymbol{\eta} = (\theta_1, \theta_2)$. Sensor j is centered at

$$s_j = 1.5(\cos\theta_j, \sin\theta_j) \tag{226}$$

and observes

$$\Phi_j(x;\theta_j) = \exp\left(-\frac{\|x - s_j\|^2}{2\ell^2}\right), \qquad \ell = 0.45. \tag{227}$$

Sensor angles are canonicalized and sorted, with minimum projective angular separation $20°$. Each finite observation trial samples $n = 100$ particles at $K = 11$ acquisition times nested in a 21-node scientific grid. At an interior acquisition time,

$$y_{k,j} = \frac{1}{100}\sum_{i=1}^{100}\Phi_j(X_i(t_k);\theta_j) + 0.01 z_{k,j}, \qquad z_{k,j} \sim \mathcal{N}(0,1). \tag{228}$$

Exact population moments are used at the two endpoints. Selection and validation use disjoint frozen common-random-number banks.

B.2.3 Frozen reference dynamics

The reference velocity is an MLP with two spatial inputs and five time features consisting of normalized time and the first two sine/cosine harmonics. It has four SiLU hidden layers of width 128 and a linear two-dimensional output. Conditional flow matching uses independent endpoint pairs with bridge

$$x_t = (1-t)x_0 + t x_1 + 0.15\sin(\pi t)z, \qquad u_t = -x_0 + x_1 + 0.15\pi\cos(\pi t)z, \qquad z \sim \mathcal{N}(0, I). \tag{229}$$

Training uses Adam for 12,000 steps, batch size 2048, gradient clipping at norm 10, and cosine learning-rate decay from 10^{-3} to 5×10^{-5}. The frozen reference is integrated through all 21 scientific time nodes using RK4 with 16 substeps per interval. The same checkpoint and reference particle bank are used for every candidate design.

B.2.4 Moment reconstruction and information projection

The eleven observations from each sensor are reconstructed with endpoint-anchored quadratic generalized least squares. The relative ridge is 10^{-12} and the variance floor is 10^{-10}. The fitted trajectory supplies $\widehat{\boldsymbol{c}}_{\boldsymbol{\eta}}(t)$ and $\dot{\widehat{\boldsymbol{c}}}_{\boldsymbol{\eta}}(t)$ at all 21 scientific time nodes.

To prevent the smoother from requesting moment vectors unsupported by the reference bank, candidate trajectories are screened against a common feasible-moment polytope. Search uses 96 directions with margin 10^{-6}, while final auditing uses exact convex-hull feasibility. At each time node, reference particles with base weights b_i are exponentially tilted according to

$$q_i(\lambda) = \frac{b_i \exp(\lambda^\top \Phi_{\boldsymbol{\eta}}(x_i))}{\sum_j b_j \exp(\lambda^\top \Phi_{\boldsymbol{\eta}}(x_j))}, \qquad \sum_i q_i(\lambda)\Phi_{\boldsymbol{\eta}}(x_i) = \widehat{\boldsymbol{c}}_{\boldsymbol{\eta}}(t). \tag{230}$$

The authoritative projection permits up to 300 iterations. Search evaluations use at most 80 iterations, residual tolerance 10^{-7}, Newton ridge 10^{-7}, step cap 20, multiplier clip 1000, six line-search reductions, and final projection acceptance tolerance 2×10^{-6}.

B.2.5 Scientific risks and transport objectives

The oracle population stage evaluates information-projected laws constructed from exact analytic moments and compares them with the hidden population. The frozen optimum and population admissibility threshold are

$$L^\star = 0.0687360118546, \qquad L_{\max} = L^\star + 0.0005 = 0.0692360118546. \tag{231}$$

The finite-data Law criterion repeats measurement reconstruction and information projection using sparse noisy measurements and evaluates Gaussian-kernel MMD with bandwidth 0.55 at the held-out scientific time nodes. The audited Law anchor is

$$R_{\text{anchor}} = 0.0661857367210. \tag{232}$$

A candidate considered by Tangent or Full must satisfy

$$L(\boldsymbol{\eta}) \le L_{\max}, \qquad R(\boldsymbol{\eta}) \le \left(1 + \frac{p}{100}\right) R_{\text{anchor}}. \tag{233}$$

For authoritative Full evaluation, bilinear deposition produces a physical raster density q_h and signed continuity forcing s_h. Both are smoothed using the same positive full-domain separable Gaussian with boundary normalization. The discrete weighted-Poisson problem is

$$-\nabla_h \cdot (q_h \nabla_h \psi_h) = s_h, \qquad \delta_h^* = -\nabla_h \psi_h, \qquad A_{\text{full},h} = \|\delta_h^*\|_{q_h}^2. \tag{234}$$

The same q_h defines both the weighted operator and the vector-field inner product. No density floor is added to the authoritative scientific operator. Conductive components are solved with a gauge constraint and a sparse direct solver, with preconditioned conjugate gradients as a fallback. The final raster is 101×101, all 21 time nodes are included, and the frozen median weighted Scott bandwidth is 0.417530106552.

B.2.6 Optimization and certification

The design workflow is Population $\to$ Law $\to$ Tangent $\to$ Full. Population minimizes L; Law minimizes finite-data R subject to the population screen; Tangent minimizes moment-tangent action under the same risk constraints; and Full minimizes the physical-density Full action.

For allowance p, define

$$\mathcal{D}_p = \left\{\boldsymbol{\eta} : L(\boldsymbol{\eta}) \le L_{\max},\ R(\boldsymbol{\eta}) \le \left(1 + \frac{p}{100}\right) R_{\text{anchor}}\right\}. \tag{235}$$

Group	Hyperparameter	Value
Global	experiment seed	20260813
Population	radius / Gaussian std. / half-width	$1.5/0.3/3.2$
Population	nuisance-angle range / quadrature	$30°$–$60°$ / 5
Measurement	sensor radius / width / min. separation	$1.5/0.45/20°$
Measurement	particles / acquisitions / noise std.	$100/11/0.01$
Reference	width / hidden layers / steps / batch	$128/4/12000/2048$
Reference	initial LR / final ratio / grad. clip	$10^{-3}/0.05/10$
Reference	bridge noise / GH order / RK4 substeps	$0.15/36/16$
Reconstruction	type / ridge / variance floor	quadratic GLS / 10^{-12} / 10^{-10}
Projection	max authoritative/search iterations	$300/80$
Projection	residual / Newton ridge / step cap	$10^{-7}/10^{-7}/20$
Projection	clip / line search / acceptance	$1000/6/2 \times 10^{-6}$
Full raster	grid / time nodes	$101 \times 101/21$
Full raster	bandwidth	0.417530106552
Full Poisson	density floor	0
Randomness	Law / Full / validation trials	$32/64/128$

Table 2: Principal numerical settings for the analytic Gaussian-mixture benchmark.

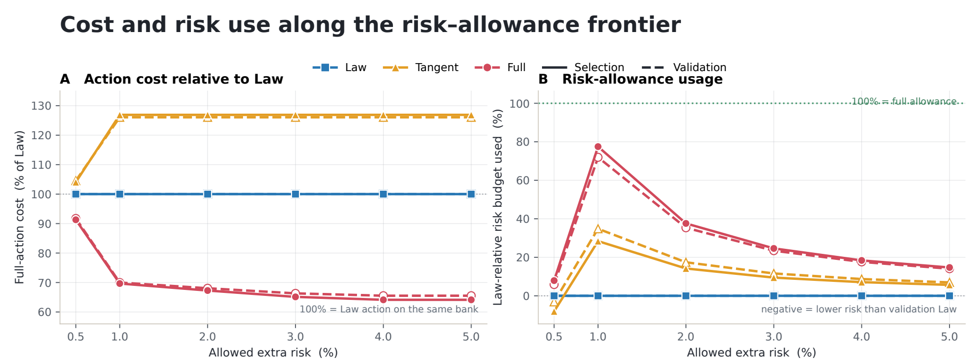

Figure 4: Risk-controlled comparison for the analytic Gaussian-mixture benchmark across the complete percentage sweep. The Full-selected geometry achieves lower common Full action than the frozen Law geometry throughout the sweep and reaches a 34.47% reduction on independent validation at the 4%–5% operating points. The Tangent-selected geometry optimizes a different, moment-rate-only objective and does not preserve the Full-law ranking. Only selection risk is constrained by the stated allowance; validation quantities are independent out-of-sample diagnostics.

Because these feasible sets are nested, the preceding certified Full winner is retained as a mandatory incumbent at the next allowance. A new candidate replaces it only when the candidate satisfies both scientific-risk screens, passes all numerical certificates, and lowers the authoritative selection action by more than 10^{-6}.

Candidate search uses a lower-cost 51×51 two-trial proxy. Leading candidates are re-evaluated at 101×101 on 12 trials, and the final winner is determined from the authoritative 101×101 evaluator on all 64 action-selection trials. The experiment uses 32 Law-selection trials, 64 Full/Tangent selection trials, and 128 independent validation trials. Validation occurs only after the geometry is frozen.

Certification checks moment calibration, effective sample size, empirical support, positive raster density, mass conservation, forcing compatibility, solver convergence, weighted-Poisson residual, Full and Tangent moment-rate feasibility, the hidden nullspace condition, Tangent–Hidden orthogonality, the Pythagorean decomposition, and the hierarchy $A_{\tan,h} \leq A_{\text{full},h}$.

Allow.	Method	Sensor angles	Risk inc.	Budget used	Sel. Full action	Certified
0.5%	Law	$23.38°, 67.95°$	0.000%	0.0%	30.202	yes
0.5%	Tangent	$24.44°, 67.28°$	-0.040%	-8.1%	31.429	yes
0.5%	Full	$23.0868°, 69.0609°$	0.040%	7.9%	27.574	yes
1%	Law	$23.38°, 67.95°$	0.000%	0.0%	30.202	yes
1%	Tangent	$25.17°, 64.15°$	0.284%	28.4%	38.325	yes
1%	Full	$20.7917°, 71.6743°$	0.775%	77.5%	21.040	yes
2%	Law	$23.38°, 67.95°$	0.000%	0.0%	30.202	yes
2%	Tangent	$25.17°, 64.15°$	0.284%	14.2%	38.325	yes
2%	Full	$21.1452°, 72.0279°$	0.752%	37.6%	20.322	yes
3%	Law	$23.38°, 67.95°$	0.000%	0.0%	30.202	yes
3%	Tangent	$25.17°, 64.15°$	0.284%	9.5%	38.325	yes
3%	Full	$21.4988°, 72.3814°$	0.739%	24.6%	19.671	yes
4%	Law	$23.38°, 67.95°$	0.000%	0.0%	30.202	yes
4%	Tangent	$25.17°, 64.15°$	0.284%	7.1%	38.325	yes
4%	Full	$21.6755°, 72.5582°$	0.735%	18.4%	19.370	yes
5%	Law	$23.38°, 67.95°$	0.000%	0.0%	30.202	yes
5%	Tangent	$25.17°, 64.15°$	0.284%	5.7%	38.325	yes
5%	Full	$21.6755°, 72.5582°$	0.735%	14.7%	19.370	yes

Table 3: Analytic Gaussian-mixture benchmark: certified selection-bank results.

Allow.	Method	Val. law risk	Val. Full action	Reduction vs. Law	Valid
0.5%	Law	0.065816 ± 0.000211	29.013980 ± 1.033466	0.00%	100%
0.5%	Tangent	0.065805 ± 0.000209	30.332182 ± 1.002152	-4.54%	100%
0.5%	Full	0.065835 ± 0.000216	26.618646 ± 0.856027	8.26%	100%
1%	Law	0.065816 ± 0.000211	29.013980 ± 1.033466	0.00%	100%
1%	Tangent	0.066045 ± 0.000208	36.576956 ± 1.537346	-26.07%	100%
1%	Full	0.066289 ± 0.000248	20.310305 ± 0.633813	30.00%	100%
2%	Law	0.065816 ± 0.000211	29.013980 ± 1.033466	0.00%	100%
2%	Tangent	0.066045 ± 0.000208	36.576956 ± 1.537346	-26.07%	100%
2%	Full	0.066281 ± 0.000250	19.746996 ± 0.542327	31.94%	100%
3%	Law	0.065816 ± 0.000211	29.013980 ± 1.033466	0.00%	100%
3%	Tangent	0.066045 ± 0.000208	36.576956 ± 1.537346	-26.07%	100%
3%	Full	0.066279 ± 0.000253	19.243082 ± 0.461129	33.68%	100%
4%	Law	0.065816 ± 0.000211	29.013980 ± 1.033466	0.00%	100%
4%	Tangent	0.066045 ± 0.000208	36.576956 ± 1.537346	-26.07%	100%
4%	Full	0.066280 ± 0.000255	19.013149 ± 0.425278	34.47%	100%
5%	Law	0.065816 ± 0.000211	29.013980 ± 1.033466	0.00%	100%
5%	Tangent	0.066045 ± 0.000208	36.576956 ± 1.537346	-26.07%	100%
5%	Full	0.066280 ± 0.000255	19.013149 ± 0.425278	34.47%	100%

Table 4: Analytic Gaussian-mixture benchmark: independent validation over 128 frozen validation trials.

B.2.7 RESULTS

The sweep uses $p \in \{0.5, 1, 2, 3, 4, 5\}\%$. The certified Full selection actions are 27.57395, 21.04026, 20.32243, 19.67140, 19.37032, and 19.37032, respectively.

Across the final Law, Tangent, and Full geometries on both selection and validation banks, the maximum raster mass error is 4.44×10^{-16}, maximum forcing compatibility error is 5.17×10^{-16}, maximum physical Poisson relative residual is 1.19×10^{-11}, maximum Full moment-rate residual is 6.62×10^{-14}, maximum Tangent moment-rate residual is 8.08×10^{-16}, maximum hidden-nullspace residual is 6.62×10^{-14}, maximum absolute Tangent–Hidden inner-product residual is 1.46×10^{-13}, and maximum absolute Pythagorean residual is 9.09×10^{-13}. The action hierarchy holds without clipping.

Allow.	$A_{\tan,h}$	$A_{\text{hid},h}$	Γ_h	**Sel. cert.**	**Val. cert.**
0.5%	0.719447	26.854504	0.973908	yes	yes
1%	0.878956	20.161304	0.958225	yes	yes
2%	0.876657	19.445775	0.956863	yes	yes
3%	0.874379	18.797024	0.955551	yes	yes
4%	0.873247	18.497072	0.954918	yes	yes
5%	0.873247	18.497072	0.954918	yes	yes

Table 5: Common-raster Full–Tangent–Hidden decomposition for the Full-selected analytic geometries.

The audited numerical Law solution should not be interpreted as a proof of the global finite-data risk optimum: the 0.5% Tangent geometry attains a risk 0.040% below the frozen Law anchor. Likewise, the multistart Full optimization certifies the declared numerical search but not global optimality over the continuous sensor-angle domain. Standard errors quantify variation across the frozen validation trials and do not include optimizer-basin, frozen-reference, or model-specification uncertainty.

B.3 Double-gyre passive-tracer benchmark

B.3.1 Hidden dynamics

The physical domain is $\Omega = [0,2] \times [0,1]$, and normalized experiment time $t \in [0,1]$ corresponds to physical time $\tau = 10t$. With $A_g = 0.1$, $\epsilon_g = 0.25$, $T_g = 10$, and $\omega = 2\pi/T_g$, define

$$a(\tau) = \epsilon_g \sin(\omega\tau), \qquad b(\tau) = 1 - 2a(\tau), \qquad f(x,\tau) = a(\tau)x_1^2 + b(\tau)x_1. \tag{236}$$

The physical-time double-gyre velocity is

$$v_1(x,\tau) = -\pi A_g \sin(\pi f)\cos(\pi x_2), \qquad v_2(x,\tau) = \pi A_g \cos(\pi f)\sin(\pi x_2)\,\partial_{x_1} f, \tag{237}$$

and the normalized-time simulator evolves according to $dX/dt = 10v$. The velocity is tangent to the rectangular boundary. The initial population contains a 10% uniform background; the remaining mass is distributed among four truncated isotropic Gaussian components in proportions $(0.30, 0.20, 0.25, 0.25)$, with standard deviation 0.07 and centers

$$(0.45, 0.25), \qquad (0.78, 0.72), \qquad (1.28, 0.28), \qquad (1.62, 0.68). \tag{238}$$

The frozen hidden-population bank contains $50{,}000$ particles; finite observation trials subsample $2{,}000$ particles.

B.3.2 Measurement model and scientific risk

A design $\boldsymbol{\eta} = (z_1, \ldots, z_4)$ consists of four sensor centers z_j in the rectangle. Sensor j measures

$$\Phi_j(x; z_j) = \exp\left(-\frac{\|x - z_j\|^2}{2(0.12)^2}\right). \tag{239}$$

Each center remains at least 0.24 from the corresponding box boundary and at least 0.24 from every other center, equivalently $x_j \in [0.24, 1.76]$, $y_j \in [0.24, 0.76]$, and $\|z_i - z_j\| \geq 0.24$. Each finite trial observes the four empirical sensor means at nine acquisition nodes nested in the 21-node scientific grid,

$$y_{k,j} = \frac{1}{2000}\sum_{i=1}^{2000} \Phi_j(X_i(t_k); z_j) + 0.005\,\xi_{k,j}, \qquad \xi_{k,j} \sim \mathcal{N}(0,1), \tag{240}$$

with exact endpoint anchoring. Bounded endpoint-anchored cubic smoothing splines reconstruct the four moment trajectories and their derivatives on all 21 scientific nodes. Scientific risk is a multiscale Gaussian MMD criterion with bandwidths $0.05, 0.10, 0.20$, and 0.40. Population feasibility is imposed before the finite-data risk cap, and Full action is never allowed to compensate for a scientifically inadmissible geometry. If R_{Law} is the frozen Law selection risk, allowance p admits only designs satisfying

$$R(\boldsymbol{\eta}) \leq \left(1 + \frac{p}{100}\right) R_{\text{Law}}. \tag{241}$$

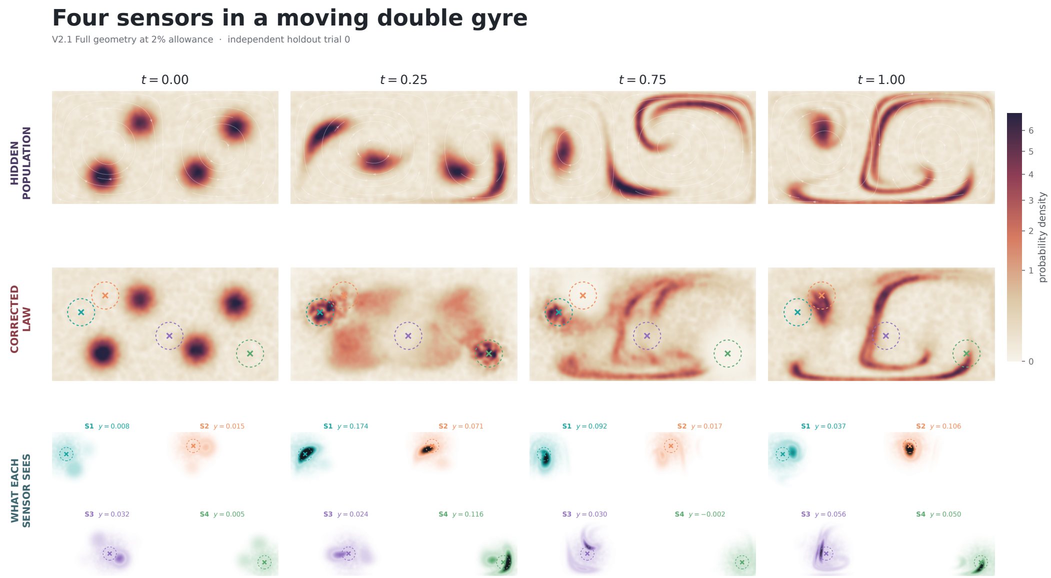

Figure 5: Representative snapshots for the confirmed 2% Full geometry. The upper row shows the hidden double-gyre population available to the benchmarker, the middle row shows the information-projected law constrained by the four reconstructed sensor moments, and the lower panels visualize the localized contributions to the four sensor readings. Agreement of the aggregate measurements does not require pointwise agreement between the hidden and projected densities.

B.3.3 Frozen references and information projection

Three independent endpoint-trained reference models are frozen before sensor selection. Each reference is a box-logit flow with a four-hidden-layer, width-128 SiLU MLP trained for 12,000 Adam steps, and each produces a 32,768-particle rollout on the common 21-node scientific grid. The physical raster bandwidth 0.058816544123815116 is fixed from the median of the three reference-only weighted Scott bandwidths and therefore does not depend on sensor geometry or action outcomes.

At each time, the empirical information projection exponentially tilts the corresponding frozen reference particle law until its four sensor moments match the reconstructed observations. If b_i are the frozen reference weights, the projected weights take the form

$$q_i(\lambda, \boldsymbol{\eta}) = \frac{b_i \exp\big(\lambda^\top \boldsymbol{\Phi}_{\boldsymbol{\eta}}(x_i)\big)}{\sum_r b_r \exp(\lambda^\top \boldsymbol{\Phi}_{\boldsymbol{\eta}}(x_r))}, \qquad \sum_i q_i(\lambda, \boldsymbol{\eta}) \boldsymbol{\Phi}_{\boldsymbol{\eta}}(x_i) = \widehat{\boldsymbol{c}}_{\boldsymbol{\eta}}(t). \tag{242}$$

The resulting distribution is a canonical completion of the four aggregate observations relative to the frozen reference, not a pointwise reconstruction of the hidden tracer density.

B.3.4 Bounded-domain Full action

The rectangular domain requires the physical density, signed continuity source, and reference flux to be treated consistently near the impermeable wall. The authoritative evaluator uses the same cell-integrated Gaussian kernel with even reflection for projected density and signed source, the corresponding reflected rule for reference flux, no artificial floor in the projected physical density, and homogeneous Neumann boundary conditions for the correction potential. Writing $K(q_h) = -\nabla_h \cdot (q_h \nabla_h)$, the discrete Full correction is obtained from

$$K(q_h)\psi_h = -s_h, \qquad \delta_h^\star = -\nabla_h \psi_h, \qquad A_{\mathrm{full},h} = \|\delta_h^\star\|_{q_h}^2. \tag{243}$$

The authoritative calculation uses a 256×128 physical grid at all 21 scientific time nodes. Evaluations fail closed unless mass conservation, source compatibility, moment calibration, covariance conditioning, effective sample size, Poisson convergence, and independent action checks all pass.

Allowance	Full sensor centers (x, y)	Sel. risk inc.	Holdout risk change	Holdout Full action	Reduction vs. Law
0.5%	$(0.251, 0.607), (0.473, 0.760), (1.061, 0.398), (1.760, 0.240)$	0.291%	0.303%	1.4598	8.28%
1%	$(0.251, 0.602), (0.468, 0.757), (1.042, 0.393), (1.760, 0.240)$	0.810%	0.850%	1.3974	12.19%
2%	$(0.257, 0.606), (0.471, 0.754), (1.038, 0.397), (1.750, 0.241)$	1.556%	1.599%	1.3414	15.71%

Table 6: Confirmed double-gyre Full designs. Selection risk is constrained by the prescribed Law-relative allowance; holdout risk is an independent cross-evaluation and does not enter selection. The common Law holdout Full action is 1.5915.

Reference	0.5% **allowance**	1% **allowance**	2% **allowance**
0	7.97% $[5.52\%, 10.43\%]$	11.81% $[9.36\%, 14.27\%]$	15.40% $[12.95\%, 17.86\%]$
1	8.30% $[5.85\%, 10.75\%]$	12.35% $[9.90\%, 14.80\%]$	15.82% $[13.37\%, 18.27\%]$
2	8.55% $[6.10\%, 11.01\%]$	12.42% $[9.97\%, 14.87\%]$	15.91% $[13.46\%, 18.37\%]$

Table 7: Reference-wise Full-versus-Law percentage reductions with simultaneous 95% intervals. Every lower bound is strictly positive. The common max-deviation half-width is 2.454 percentage points, and the maximum within-reference relative standard error is 2.442%.

B.3.5 Selection and independent confirmation

Candidate generation uses one frozen 128-trial selection bank shared across all methods and all three references. Selection is feasibility-first: the population screen and exact finite-data risk cap determine the admissible set before Full action is used for ranking. Promoted candidates are then evaluated with the authoritative 256×128 solver. The confirmed Full geometries are selected at $p \in \{0.5, 1, 2\}$.

After selection, one Law geometry and the three Full geometries are frozen and evaluated on a fresh shared 64-trial holdout under each of the three references. This produces $3 \times 4 \times 64 = 768$ primary exact reference/design/trial evaluations. The primary inferential family consists of the nine reference-by-allowance Full-versus-Law reductions and uses common max-deviation simultaneous 95% intervals. Tangent geometries are also evaluated on the same holdout as a supplementary descriptive comparison, but they are not included in the primary simultaneous inference family.

B.3.6 Results

The Full reduction increases across the evaluated range from 8.28% to 15.71%. More importantly, the effect is reproduced under each independently trained reference. The simultaneous reference-wise intervals are

The independently evaluated Tangent geometries reduce holdout Full action by 5.21%, 5.85%, and 3.45% at the 0.5%, 1%, and 2% allowances, respectively. Thus Tangent improves over Law under the common Full metric but remains weaker than Full at every evaluated allowance, with its improvement decreasing at 2% while the Full reduction continues to increase. This provides a direct finite-sample illustration of the distinction between minimizing correction visible through the measured moment rates and minimizing the correction required by the complete projected law.

All 768 primary exact holdout evaluations pass the frozen numerical gates. The reported Full-versus-Law result therefore combines feasibility-first selection, a fresh shared holdout, three independently trained frozen references, and a bounded-domain Full-action discretization that treats density, source, flux, and boundary conditions consistently.

B.4 Prospective aggregate-only double-gyre benchmark

B.4.1 Information boundary and physical setup

The prospective experiment uses the same physical double-gyre system, initial population, four-sensor geometry, and finite observation model as the retrospective benchmark in Appendix B.3. In particular, the state lies in $[0, 2] \times [0, 1]$, each Gaussian sensor has width 0.12, sensor centers obey the same 0.24 boundary and pairwise-separation constraints, and a finite trial corresponds to 2,000

Allowance	D0 Law risk	D0 Full risk	D0 ceiling	D0 Law action	D0 Full action	Selection outcome
0.5%	1.19657794	1.19657796	1.20256083	1.55723989	1.55723989	Law incumbent
1%	1.19657794	1.19657796	1.20854372	1.55723989	1.55723989	carried 0.5% incumbent
2%	1.19657794	1.21481697	1.22050950	1.55723989	1.29078832	audited feasible finalist

Table 8: Repaired D0 selection frontier for the prospective double-gyre experiment. The 0.5% and 1% Full solutions coincide with Law; the distinct 2% finalist remains inside the repaired risk ceiling.

particles observed at nine acquisition nodes with additive detector-noise standard deviation 0.005. The reconstructed moment path is evaluated on the same 21-node scientific grid.

The distinction is informational rather than physical. Before sensor selection, the prospective target interface exposes only the two endpoint ensembles, aggregate sensor-response mean and second-moment fields on a fixed physical grid, analytically recovered cross-sensor second moments sufficient to determine the complete finite-sampling covariance, and five aggregate scientific-QoI prediction curves with frozen scales. It exposes no intermediate target particles, simulator object, hidden-state path, or loader into the hidden-validation tree. The $50{,}000$-particle hidden trajectory is sealed during selection and is used only after all three allowance-specific geometries have been frozen.

B.4.2 Reference roles, moment reconstruction, and Full action

A single endpoint-trained reference, denoted D0, is frozen for design. Its 32,768-particle rollout provides the common selection-stage background law and velocity. At each time, the four reconstructed sensor moments define a moment fiber and the D0 marginal is exponentially tilted to match those moments, exactly as in the empirical projection construction of Appendix B.3. Endpoint-anchored cubic splines reconstruct the four moment trajectories on the 21 scientific nodes. A smooth C^2 bounded transformation keeps reconstructed moments inside the feasible interval without the zero-gradient plateaus introduced by hard clipping.

The prospective Full-action evaluator uses a matched reflected bounded-domain discretization. Projected density and signed continuity source are constructed with the same direct cell-integrated even-reflection Gaussian operator, and the reference flux uses the corresponding reflected convention. No artificial floor is added to the physical projected density. With

$$K(q_h) = -\nabla_h \cdot (q_h \nabla_h), \tag{244}$$

the discrete correction again solves

$$K(q_h)\psi_h = -s_h, \qquad \delta_h^\star = -\nabla_h \psi_h, \qquad A_{\mathrm{full},h} = \|\delta_h^\star\|_{q_h}^2, \tag{245}$$

with homogeneous Neumann boundary conditions. Authoritative prospective evaluation uses a 128×64 grid at all 21 scientific time nodes. The Gaussian bandwidth is fixed from the frozen reference only and does not depend on grid spacing, sensor geometry, trial, risk, or action. The evaluator fails closed on the prescribed geometry, risk, projection-calibration, covariance, effective-sample-size, Poisson, moment-rate, compatibility, and finite-value checks.

B.4.3 Repaired Law anchor and one-way freeze protocol

The prospective study uses the same percentage-risk rule equation 222 with $p \in \{0.5, 1, 2\}$. A stronger D0-only risk audit found that the originally frozen Law baseline was underoptimized: the repaired Law risk is

$$R_{\mathrm{Law}}^{\mathrm{D0}} = 1.19657794045, \tag{246}$$

which is 2.4957% below the superseded value 1.22720547227. The originally selected Full geometries therefore no longer satisfied all repaired allowance ceilings. Full was rerun at 0.5% and 1%; at both tight allowances the certified minimum-action choice was the repaired Law geometry itself. At 2%, a previously audited authoritative finalist remained feasible under the repaired ceiling and was retained.

The repaired D0 selection frontier is

The final freeze order is one-way. Aggregate target fields and endpoints are fixed first; the byte-identical D0 reference is reused; Law is reoptimized using D0 only; Full is rerun at 0.5% and 1%

Allowance	E1 Law action	E1 Full action	Reduction	E1 Law risk	E1 Full risk	Paired 95% CI	Result
0.5%	1.70295372	1.70295372	0.00%	1.15611926	1.15611926	$[0, 0]$	no benefit
1%	1.70295372	1.70295372	0.00%	1.15611926	1.15611926	$[0, 0]$	no benefit
2%	1.70295372	1.42699969	16.20%	1.15611926	1.17101696	$[-0.31510, -0.23681]$	pass

Table 9: Fresh E1 validation for the repaired prospective frontier. The paired confidence interval is for the trial-wise Full-minus-Law action difference. At 2%, the observed E1 risk increase is 1.2886%, below the prescribed 2% allowance, and the paired interval lies strictly below zero.

and the audited 2% finalist is adopted; all three points are frozen; only then is a fresh independent endpoint reference E1 trained and the fresh hidden-validation states and observation randomness generated. E1 is validation-only and is never folded back into the design objective. The labels D0 and E1 denote optimization artifacts, not different physical regimes.

B.4.4 Post-freeze E1 validation

After the repaired three-point frontier was frozen, the independent E1 reference and 64 fresh paired hidden trials produced

Thus the repaired 0.5% and 1% points certify feasibility but no improvement over Law. At 2%, the prospectively selected geometry transfers to the post-freeze E1 reference and lowers held-out Full action by 16.20%. Tangent is not reported for the repaired prospective frontier because it was not rerun after the Law repair; any Law geometry stored in a Tangent slot for validation-cache reuse is only a placeholder and is not a Tangent result.

The prospective uncertainty statement is intentionally narrower than the retrospective three-reference result. There is one D0 design-reference seed and one independent E1 validation-reference seed. The 64 paired trials quantify observation randomness conditional on E1; they do not establish robustness across reference-model seeds. The prospective and retrospective double-gyre studies should therefore be compared structurally, as tests under different pre-selection information sets, rather than treated as replicate estimates of a common effect. No claim is made for allowances above 2%.

B.5 Hardware Configuration

Experiments were executed on a high-performance laptop equipped with an Intel Core Ultra 9 275HX processor, comprising 24 physical cores and supporting a maximum turbo frequency of 5.4 GHz. The high-performance laptop also contained an NVIDIA GeForce RTX 5090 Laptop GPU with 24 GB of dedicated GDDR7 memory.

The computational workload was divided between the CPU and GPU. JAX-based particle simulations, reference-flow training, and automatic differentiation were accelerated using the NVIDIA GPU. Batched empirical information-projection trajectories were evaluated using a native C++17 implementation parallelized over the CPU with OpenMP.

B.6 Software Environment

The computational environment was hosted in Windows Subsystem for Linux 2 (WSL2), running Ubuntu 24.04.3 LTS with Linux kernel 6.6.87.2-microsoft-standard-WSL2. The principal implementation used Python 3.12.3 and JAX 0.8.3 with 64-bit numerical precision enabled. GPU acceleration was provided through the JAX CUDA 12 backend.

The information-projection stage used the Tesseract runtime to execute an in-process native C++ extension. The extension evaluates complete batched empirical information-projection trajectories using a double-precision Newton solver. Multiplier estimates are warm-started between consecutive time nodes, while independent trajectories and linear-algebra operations are parallelized using OpenMP. The native solver is exposed to Python through pybind11 and integrated with JAX through `tesseract-jax`. Jacobian–vector and vector–Jacobian products are computed using implicit moment-covariance solves rather than by differentiating through the individual Newton iterations.

The C++ extension was compiled in release mode using the GNU C++ 13.3 compiler, the C++17 language standard, and the optimization flags `-O3` and `-march=native`. Fast-math optimizations were not enabled in order to preserve the numerical behavior of the double-precision projection solver. The complete software environment is summarized in Table 10.

Table 10: Hardware and software environment used for the experiments.

Component	**Configuration**
Operating system	Ubuntu 24.04.3 LTS under WSL2
Linux kernel	6.6.87.2-microsoft-standard-WSL2
Processor	Intel Core Ultra 9 275HX (24 cores, up to 5.4 GHz)
GPU	NVIDIA GeForce RTX 5090 Laptop GPU
GPU memory	24 GB GDDR7
NVIDIA driver	581.57
Python	3.12.3
JAX	0.8.3
jaxlib	0.8.3
Accelerator backend	CUDA 12 via the JAX CUDA plugin 0.8.3
CUDA runtime	12.9 (JAX-packaged runtime)
NumPy	2.5.1
SciPy	1.18.0
Tesseract Core	`tesseract-core` 1.11.0
Tesseract JAX interface	`tesseract-jax` 0.2.3
Projection backend	Native Tesseract C++/OpenMP batched trajectory solver
C++ language standard	C++17
C/C++ compiler	GCC/G++ 13.3.0
Build system	CMake 4.4.2, release configuration
Python/C++ bindings	pybind11 3.1.0
CPU parallelization	OpenMP 4.5
Native compilation	`-O3`, `-march=native`, without `-ffast-math`
Native numerical precision	IEEE 754 double precision (`float64`)
Neural optimization	Custom JAX implementation of Adam
JAX numerical precision	64-bit floating point enabled
Job scheduling	Local execution; no external scheduler